DMA002A
Mathematics-II

Department of Mathematics

JECRC University Jaipur

Unit-III
Ordinary Differential Equations of Higher Orders

DMA025A	**Mathematics-II**	**3+1+0 = 4 Credits**

Course Contents:

UNIT I: Matrices:
Linear Systems of Equations, Linear Independence, Rank of a Matrix; Determinant, Inverse of a matrix, rank-nullity theorem, System of linear equations, Symmetric, skew-symmetric and orthogonal matrices, Determinants, Eigen values and eigenvectors, Orthogonal transformation, Diagonalization of matrices, Cayley-Hamilton Theorem.

UNIT II: First order ordinary differential equations:
Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type.

UNIT III: Ordinary differential equations of higher orders:
Second order linear differential equations with variable coefficients: Euler-Cauchy equations, solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.

UNIT IV: Complex Variable–Differentiation:

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate, elementary analytic functions(exponential, trigonometric, logarithm) and their properties, Conformal mappings, Mobius transformations and their properties.

UNIT V: Complex Variable–Integration:
Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour.

Text/Reference Books:

- AICTE's Prescribed Textbook: Mathematics-II (Calculus, Ordinary Differential Equations and Complex Variable), Khanna Book Publishing Co.
- Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
- Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
- Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons, 2006.
- Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.

Course Outcome (CO):

JECRC UNIVERSITY
BUILD YOUR WORLD

Upon successful completion of this course, the student will be able to:

- CO1: The students will learn the essential tool of matrices and linear algebra in a comprehensive manner.
- CO2: Understand the linear ordinary differential equations along with their applications.
- CO3: Will able to solve problem involving ordinary differential equations of higher orders.
- CO4: The students will learn the Complex Variable – Differentiation.
- CO5: The students will learn the Complex Variable–Integration.

Mapping of COs with Pos

MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course Outcome	Program Outcome												Program Specific Outcome		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	M	L			M		M					M		M	
CO2	M	M		L	M		M			M	M	M	M		
CO3		L			M					L		M	M	M	
CO4	M	M		M	L		M					M		M	
CO5	M			M	M		H					M	M	M	

H=Highly Related ;M=Medium L=Low

Contents

- Second order linear differential equations with variable coefficients: Euler-Cauchy equations,
- Solution by variation of parameters;
- Power series solutions: Legendre's equations and Legendre polynomials,
- Frobenius method,
- Bessel's equation and Bessel's functions of the first kind and their properties.

LINEAR DIFFERENTIAL EQUATIONS

If the degree of the dependent variable and all derivatives is one, such differential equations are called *linear differential equations e.g.*

(1) $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y=x^2+x+1$ (2) $2\frac{d^2x}{dt^2}-\frac{dx}{dt}-3x=f(t)$

NON LINEAR DIFFERENTIAL EQUATIONS

If the degree of the dependent variable and / or its derivatives are of greater than 1 such differential equations are called one-linear differential equations.

(1) $\frac{d^2y}{dx^2}+\frac{dy}{dx}+y^2=\sin x$ (2) $\frac{d^2y}{dx^2}+2\left(\frac{dy}{dx}\right)^2+y^2=e^x$ (3) $\left(\frac{d^2x}{dt^2}\right)^2+\frac{dx}{dt}+x=f(t)$

The order of a differential equation is the highest order of the derivative involved. All the above differential equations are of second order.

Fourier and Laplace transforms are mathematical tools to solve the differential equations.

LINEAR DIFFERENTIAL EQUATIONS OF SECOND ORDER WITH CONSTANT COEFFICIENTS

The general form of the linear differential equation of second order is

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R$$

where P and Q are constants and R is a function of x or constant.

Differential operator. Symbol D stands for the operation of differential *i.e.*,

$$Dy = \frac{dy}{dx}, \quad D^2y = \frac{d^2y}{dx^2}$$

$\frac{1}{D}$ stands for the operation of integration.

$\frac{1}{D^2}$ stands for the operation of integration twice.

$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R$ can be written in the operator form.

$$D^2y + P\,Dy + Q\,y = R \qquad \Rightarrow \qquad (D^2 + PD + Q)\,y = R$$

$\Rightarrow \quad \frac{dv}{dx} + Pv = Q$ which shows that $y = v$ is the solution of $\boxed{\frac{dy}{dx} + Py = Q}$

Solution of the differential equation (1) is (2) consisting of two parts *i.e.* cu and v. cu is the solution of the differential equation whose R.H.S. is zero. cu is known as *complementary function*. Second part of (2) is v free form any arbitrary constant and is known as *particular integral.*

Complete Solution = Complementary Function + Particular Integral.

$$\Rightarrow \quad \boxed{y = C.F. + P.I.}$$

METHOD FOR FINDING THE COMPLEMENTARY FUNCTION

(1) In finding the complementary function, R.H.S. of the given equation is replaced by zero.

(2) Let $y = C_1 e^{mx}$ be the C.F. of

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0 \qquad ...(1)$$

Putting the values of y, $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in (1) then $C_1 e^{mx}(m^2 + Pm + Q) = 0$

$\Rightarrow \quad m^2 + Pm + Q = 0$. It is called **Auxiliary equation**.

(3) Solve the auxiliary equation :

Case I : Roots, Real and Different. If m_1 and m_2 are the roots, then the C.F. is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

Case II : Roots, Real and Equal. If both the roots are m_1, m_1 then the C.F. is

$$y = (C_1 + C_2 x)\, e^{m_1 x}$$

Example 1. *Solve:* $\dfrac{d^2 y}{dx^2} - 8\dfrac{dy}{dx} + 15y = 0.$

Solution. Given equation can be written as

$$(D^2 - 8D + 15)\, y = 0$$

Here auxiliary equation is $m^2 - 8m + 15 = 0$

$\Rightarrow \qquad (m - 3)(m - 5) = 0 \qquad \therefore\ m = 3, 5$

Hence, the required solution is

$$y = C_1 e^{3x} + C_2 e^{5x}$$ **Ans.**

Example 2. *Solve*: $\dfrac{d^2 y}{dx^2} - 6\dfrac{dy}{dx} + 9y = 0$

Solution. Given equation can be written as

$$(D^2 - 6D + 9)\, y = 0$$

A.E. is $m^2 - 6m + 9 = 0 \quad \Rightarrow \quad (m - 3)^2 = 0 \quad \Rightarrow \quad m = 3, 3$

Hence, the required solution is

$$y = (C_1 + C_2 x)\, e^{3x}$$ **Ans.**

Example 4. *The general solution of the differential equation*

$$\frac{d^5y}{dx^5}-\frac{d^3y}{dx^3}=0 \text{ is given by} \qquad (U.P.\ II\ Semester,\ 2009)$$

Solution. Here, we have

$$\frac{d^5y}{dx^5}-\frac{d^3y}{dx^3}=0$$

or $\quad D^5y - D^3y = 0 \quad \Rightarrow \quad (D^5 - D^3)y = 0 \quad \Rightarrow \quad D^3(D^2 - 1)y = 0$

A.E. is $m^3(m^2 - 1) = 0 \quad \Rightarrow \quad m = 0, 0, 0, 1, -1$

Here the solution is

$$y=(C_1+C_2x+C_3x^2)+C_3e^x+C_5e^{-x}$$ **Ans.**

Case III: Roots Imaginary. If the roots are $\alpha \pm i\beta$, then the solution will be

$$y=C_1e^{(\alpha+i\beta)x}+C_2e^{(\alpha-i\beta)x}=e^{\alpha x}.[C_1e^{i\beta x}+C_2e^{-i\beta x}]$$
$$= e^{\alpha x}[C_1(\cos\beta x+i\sin\beta x)+C_2(\cos\beta x-i\sin\beta x)]$$
$$= e^{\alpha x}[(C_1+C_2)\cos\beta x+i(C_1-C_2)\sin\beta x]$$
$$= e^{\alpha x}[A\cos\beta x+B\sin\beta x]$$

EXERCISE

Solve the following equations :

1. $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$ **Ans.** $y = C_1 e^x + C_2 e^{2x}$

2. $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 30y = 0$ **Ans.** $y = C_1 e^{5x} + C_2 e^{-6x}$

3. $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 0$ **Ans.** $y = (C_1 + C_2 x)\, e^{4x}$

4. $\frac{d^2y}{dx^2} + \mu^2 y = 0$ **Ans.** $y = C_1 \cos \mu x + C_2 \sin \mu x$

5. $(D^2 + 2D + 2)\, y = 0,\ y(0) = 0,\ y'(0) = 1$ (*A.M.I.E.T.E., June 2006*) **Ans.** $y = e^{-x} \sin x$

6. $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 8y = 0$ **Ans.** $y = C_1 e^{2x} + C_2 \cos 2x + C_3 \sin 2x$

7. $\frac{d^4y}{dx^4} - 32\frac{d^2y}{dx^2} + 256 = 0$ (*A.M.I.E.T.E., Dec. 2004*) **Ans.** $y = (C_1 + x)\cos 4x + (C^3 + C^4 x)\sin 4x$

8. $\frac{d^4y}{dx^4} - 4\frac{d^3y}{dx^3} + 8\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 4y = 0$ **Ans.** $y = e^x\,[(C_1 + C_2 x)\cos x + (C_3 + C_4 x)\sin x]$

RULES TO FIND PARTICULAR INTEGRAL

(i) $\dfrac{1}{f(D)}e^{ax} = \dfrac{1}{f(a)}e^{ax}$

If $f(a) = 0$ then $\dfrac{1}{f(D)}\cdot e^{ax} = x\cdot\dfrac{1}{f'(a)}\cdot e^{ax}$

If $f'(a) = 0$ then $\dfrac{1}{f(D)}\cdot e^{ax} = x^2\dfrac{1}{f''(a)}\cdot e^{ax}$

(ii) $\dfrac{1}{f(D)}x^n = [f(D)]^{-1}x^n$ Expand $[f(D)]^{-1}$ and then operate.

(iii) $\dfrac{1}{f(D^2)}\sin ax = \dfrac{1}{f(-a^2)}\sin ax$ and $\dfrac{1}{f(D^2)}\cos ax = \dfrac{1}{f(-a^2)}\cos ax$

If $f(-a^2) = 0$ then $\dfrac{1}{f(D^2)}\sin ax = x\cdot\dfrac{1}{f'(-a^2)}\cdot\sin ax$

(iv) $\dfrac{1}{f(D)}e^{ax}\cdot\phi(x) = e^{ax}\cdot\dfrac{1}{f(D+a)}\phi(x)$

(v) $\dfrac{1}{D+a}\phi(x) = e^{-ax}\int e^{ax}\cdot\phi(x)\,dx$

Example 6. *Solve* : $\frac{d^2y}{dx^2}+6\frac{dy}{dx}+9y=5e^{3x}$

Solution. $(D^2 + 6D + 9)y = 5e^{3x}$

Auxiliary equation is $m^2 + 6m + 9 = 0 \quad \Rightarrow \quad (m + 3)^2 = 0 \quad \Rightarrow \quad m = -3, -3,$

$$\text{C.F.} = (C_1 + C_2x)\, e^{-3x}$$

$$\text{P.I.} = \frac{1}{D^2+6D+9}.5.e^{3x} = 5\frac{e^{3x}}{(3)^2+6(3)+9} = \frac{5e^{3x}}{36}$$

The complete solution is $y=(C_1+C_2x)e^{-3x}+\frac{5e^{3x}}{36}$ **Ans.**

Example 7. *Solve* : $$\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=6e^{3x}+7e^{-2x}-\log 2$$

Solution. $$(D^2-6D+9)y=6e^{3x}+7e^{-2x}-\log 2$$

A.E. is $(m^2-6m+9)=0 \quad \Rightarrow \quad (m-3)^2=0, \quad \Rightarrow \quad m=3,3$

$$\text{C.F.}=(C_1+C_2x)e^{3x}$$

$$\text{P.I.}=\frac{1}{D^2-6D+9}6e^{3x}+\frac{1}{D^2-6D+9}7e^{-2x}+\frac{1}{D^2-6D+9}(-\log 2)$$

$$=x\frac{1}{2D-6}6e^{3x}+\frac{1}{4+12+9}7e^{-2x}-\log 2\frac{1}{D^2-6D+9}e^{0x}$$

$$=x^2\frac{1}{2}\cdot 6\cdot e^{3x}+\frac{7}{25}e^{-2x}-\log 2\left(\frac{1}{9}\right)=3x^2e^{3x}+\frac{7}{25}e^{-2x}-\frac{1}{9}\log 2$$

Complete solution is $y=(C_1+C_2\,x)e^{3x}+3x^2e^{3x}+\frac{7}{25}e^{-2x}-\frac{1}{9}\log 2$ **Ans.**

EXERCISE

JECRC UNIVERSITY
BUILD YOUR WORLD

Solve the following differential equations:

1. $[D^2 + 5D + 6]\,[y] = e^x$ **Ans.** $C_2\, e^{-2x} + C_2 e^{-3x} + \dfrac{e^x}{12}$

2. $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = e^{3x}$ **Ans.** $C_1 e^x + C_2 e^{2x} + \dfrac{e^{3x}}{2}$
(A.M.I.E.T.E. June 2010, 2007)

3. $(D^3 + 2D^2 - D - 2)\, y = e^x$ **Ans.** $C_1 e^x + C_2 e^{-x} + C_3 e^{-2x} + \dfrac{x}{6} e^x$

4. $\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} + 2y = \sinh x$ **Ans.** $e^{-x}[C_1 \cos x + C_2 \sin x] + \dfrac{e^x}{10} - \dfrac{e^{-x}}{2}$

5. $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 5y = -2\cosh x$ **Ans.** $e^{-2x}(C_1 \cos x + C_2 \sin x) - \dfrac{1}{10} e^x - \dfrac{e^{-x}}{2}$

6. $(D^3 - 2D^2 - 5D + 6)\, y = e^{3x}$ **Ans.** $C_1 e^x + C_2 e^{-2x} + C_3 e^{3x} + \dfrac{x.e^{3x}}{10}$

7. $\dfrac{d^3y}{dx^3} - \dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} - 4y = e^x$ **Ans.** $C_1 e^x + C_2 \cos 2x + C_3 \sin 2x + \dfrac{x\, e^x}{5}$

8. $\dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} + 9y = e^{3x}$ **Ans.** $(C_1 + C_2 x) e^{3x} + \dfrac{x^2}{2} e^{3x}$

9. $\dfrac{d^3y}{dx^3} + 3\dfrac{d^2y}{dx^2} + 3\dfrac{dy}{dx} + y = e^{-x}$ **Ans.** $(C_1 + C_2 x + C_3 x^2) e^{-x} + \dfrac{x^3}{6} e^{-x}$

10. $\dfrac{d^2y}{dx^2} - \dfrac{dy}{dx} - 6y = e^x \cosh 2x$ **Ans.** $C_1 e^{3x} + C_2 e^{-2x} + \dfrac{1}{10} x e^{3x} - \dfrac{1}{8} e^{-x}$

$$\boxed{\frac{1}{f(D^2)}\sin ax = \frac{\sin ax}{f(-a^2)}}$$

$$\boxed{\frac{1}{f(D^2)}\cdot\cos ax = \frac{\cos ax}{f(-a^2)}}$$

$D(\sin ax) = a.\cos ax$, $D^2(\sin ax) = D(a\cos ax) = -a^2.\sin ax$

$D^4(\sin ax) = D^2.D^2(\sin ax) = D^2(-a^2\sin ax) = (-a^2)^2\sin ax$

$(D^2)^n \sin ax = (-a^2)^n \sin ax$

Hence, $f(D^2)\sin ax = f(-a^2)\sin ax$

$$\frac{1}{f(D^2)}\cdot f(D^2)\sin ax = \frac{1}{f(D^2)}.f(-a^2).\sin ax$$

$$\sin ax = f(-a^2)\frac{1}{f(D^2)}\sin ax \quad \Rightarrow \quad \frac{1}{f(D^2)}\cdot\sin ax = \frac{\sin ax}{f(-a^2)}$$

Similarly, $$\frac{1}{f(D^2)}\cos ax = \frac{\cos ax}{f(-a^2)}$$

If $f(-a^2) = 0$ then above rule fails.

$$\frac{1}{f(D^2)}\sin ax = x\frac{\sin ax}{f'(-a^2)}$$

If $f'(-a^2) = 0$ then, $$\frac{1}{f(D^2)}\sin ax = x^2\frac{\sin ax}{f''(-a^2)}$$

Example 9. *Solve* : $(D^2 + 4)\, y = \cos 2x$

(*R.G.P.V., Bhopal June, 2008, A.M.I.E.T.E. Dec 2008*)

Solution. $(D^2 + 4)\, y = \cos 2x$

Auxiliary equation is $m^2 + 4 = 0$

$m = \pm 2i,\qquad \text{C.F.} = A\cos 2x + \text{B}\sin 2x$

$$\text{P.I.} = \frac{1}{D^2+4}\cos 2x = x.\frac{1}{2D}\cos 2x = \frac{x}{2}\left(\frac{1}{2}\sin 2x\right) = \frac{x}{4}\sin 2x$$

Complete solution is $y = A\cos 2x + B\sin 2x + \dfrac{x}{4}\sin 2x$ **Ans.**

Example 10. *Solve* : $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^3} + 4\frac{dy}{dx} - 2y = e^x + \cos x$ (*U.P., II Semester, Summer* 2006, 2001)

Solution. Given $(D^3 - 3D^2 + 4D - 2)\, y = e^x + \cos x$

A.E. is $m^3 - 3m^2 + 4m - 2 = 0$

$\Rightarrow \quad (m-1)(m^2 - 2m + 2) = 0$, *i.e.*, $m = 1,\ 1 \pm i$

$\therefore \quad$ C.F. $= C_1 e^x + e^x (C_2 \cos x + C_3 \sin x)$

$$\text{P.I.} = \frac{1}{(D-1)(D^2-2D+2)} e^x + \frac{1}{D^3 - 3D^2 + 4D - 2} \cos x$$

$$= \frac{1}{(D-1)(1-2+2)} e^x + \frac{1}{(-1)D - 3(-1) + 4D - 2} \cos x$$

$$= \frac{1}{(D-1)} e^x + \frac{1}{3D+1} \cos x = x\,\frac{1}{1} e^x + \frac{3D-1}{9D^2-1} \cos x$$

$$= e^x . x + \frac{(-3\sin x - \cos x)}{-9-1} = e^x . x + \frac{1}{10}(3\sin x + \cos x)$$

Hence, complete solution is

$$y = C_1 e^x + e^x (C_2 \cos x + C_3 \sin x) + x\, e^x + \frac{1}{10}(3\sin x + \cos x) \qquad \textbf{Ans.}$$

EXERCISE

Solve the following differential equations :

1. $\dfrac{d^2y}{dx^2}+6y=\sin 4x$ **Ans.** $C_1\cos\sqrt{6}x+C_2\sin\sqrt{6}x-\dfrac{1}{10}\sin 4x$

2. $\dfrac{d^2x}{dt^2}+2\dfrac{dx}{dt}+3x=\sin t$ **Ans.** $e^{-t}[A\cos\sqrt{2}t+B\sin\sqrt{2}t]-\dfrac{1}{4}(\cos t-\sin t)$

3. $\dfrac{d^2x}{dt^2}+2\dfrac{dx}{dt}+5x=\sin 2t$, given that when $t=0$, $x=3$ and $\dfrac{dx}{dt}=0$

Ans. $e^{-t}\left[\dfrac{55}{17}\cos 2t+\dfrac{53}{34}\sin 2t\right]-\dfrac{1}{17}(4\cos 2t-\sin 2t)$

4. $\dfrac{d^2y}{dx^2}-7\dfrac{dy}{dx}+6y=2\sin 3x$, given that $y=1$, $\dfrac{dy}{dx}=0$ when $x=0$.

Ans. $-\dfrac{13}{75}e^{6x}+\dfrac{27}{25}e^{x}+\dfrac{1}{75}(7\cos 3x-\sin 3x)$

5. $(D^3+1)\,y=2\cos^2 x$

Ans. $C_1e^{-x}+e^{\frac{1}{2}x}\left(C_2\cos\dfrac{\sqrt{3}}{2}x+C_3\sin\dfrac{\sqrt{3}}{2}x\right)+1+\dfrac{1}{65}(-8\sin 2x+\cos 2x)$

6. $(D^2+a^2)\,y=\sin ax$ *(A.M.I.E.T.E., June 2009)* **(Ans.** $C_1\cos ax+C_2\sin ax-\dfrac{x}{2a}\cos ax$

7. $(D^4+2a^2D^2+a^4)\,y=8\cos ax$ **Ans.** $(C_1+C_2x+C_3\cos ax+C_4\sin ax)-\dfrac{x^2}{a^2}\cos ax$

8. $\dfrac{d^2y}{dx^2}+3\dfrac{dy}{dx}+2y=\sin 2x$ *(A.M.I.E.T.E., Summer 2002)*

P.I. of the $\boxed{\dfrac{1}{f(D)}x^n = [f(D)]^{-1}x^n.}$

JECRC UNIVERSITY
BUILD YOUR WORLD

Q. Solve $\dfrac{d^3y}{dx^3} - \dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} = 1+x^2$

Soln. In operator form, $(D^3 - D^2 - 6D)y = 1+x^2$

Auxiliary Equation is, $m^3 - m^2 - 6m = 0$

$\Rightarrow m(m^2 - m - 6) = 0$

$\Rightarrow m(m-3)(m+2) = 0 \Rightarrow m = 0, -2, 3$

C.F. $= C_1 + C_2 e^{-2x} + C_3 e^{3x}$.

$$P.I. = \frac{1}{D(D+2)(D-3)}(1+x^2) = \frac{1}{D^3 - D^2 - 6D}(1+x^2) = -\frac{1}{6D}\cdot\frac{1}{\left(1 - \frac{D^2 - D}{6}\right)}(1+x^2)$$

JECRC Foundation

$$= -\frac{1}{6D}\left[1-\frac{D^2-D}{6}\right]^{-1}(1+x^2)$$

$$= -\frac{1}{6D}\left[1+\frac{D^2-D}{6}+\frac{(D^2-D)^2}{36}\right](1+x^2)$$

$$= -\frac{1}{6D}\left[(1+x^2)+\left(\frac{D^2-D}{6}\right)(1+x^2)+\frac{(D^4+D^2-2D^3)}{36}(1+x^2)\right]$$

$$= -\frac{1}{6D}\left[(1+x^2) - \frac{D}{6}(1+x^2)+\frac{7D^2}{36}(1+x^2)\right]$$

$$= -\frac{1}{6D}\left[(1+x^2)-\frac{1}{3}x+\frac{7}{18}\right] = -\frac{1}{6D}\left[x^2-\frac{1}{3}x+\frac{25}{18}\right]$$

$$= -\frac{1}{6}\left[\frac{x^3}{3}-\frac{x^2}{6}+\frac{25}{18}x\right] = -\frac{1}{108}\left[6x^3-3x^2+25x\right]$$

general solution is

$$y = C.F. + P.I. = c_1 + c_2 e^{-2x} + c_3 e^{3x} - \frac{1}{108}\left[6x^3-3x^2+25x\right]$$ Ans

EXERCISE

Solve the following equations :

1. $(D^2 + 5D + 4)\, y = 3 - 2x$ **Ans.** $C_1 e^{-x} + C_2 e^{-4x} + \frac{1}{8}(11 - 4x)$

2. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = x$ **Ans.** $(C_1 + C_2 x)\, e^{-x} + x - 2$

3. $(2D^2 + 3D + 4)\, y = x^2 - 2x$ **Ans.** $e^{-\frac{3}{4}x}[A\cos\frac{\sqrt{23}}{4}x + B\sin\frac{\sqrt{23}}{4}x] + \frac{1}{32}[8x^2 - 28x + 13]$

4. $(D^2 - 4D + 3)\, y = x^3$ **Ans.** $C_1 e^{x} + C_2 e^{3x} + \frac{1}{27}(9x^3 + 36x^2 + 78x + 80)$.

5. $5\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} - 6\frac{dy}{dx} = 1 + x^2$. **Ans.** $A + Be^{-2x} + C\,e^{3x} - \frac{1}{36}\left(2x^3 - x^2 + \frac{25}{3}x\right)$

6. $\frac{d^4y}{dx^4} + 4y = x^4$ **Ans.** $e^{x}(C_1\cos x + C_2\sin x) + e^{-x}(C_3\cos x + C_4\sin x) + \frac{1}{4}(x^4 - 6)$

7. $\frac{d^2y}{dx^2} + 2p\frac{dy}{dx} + (p^2 + q^2)y = e^{cx} + p.q\, x^2$

Ans. $e^{-px}[C_1\cos qx + C_2\sin qx] + \frac{e^{Cx}}{(p+C)^2 + q^2} + \frac{pq}{p^2+q^2}\left[x^2 - \frac{4px}{p^2+q^2} + \frac{6p^2 - 2q^2}{(p^2+q^2)^2}\right]$

8. $D^2(D^2 + 4)\, y = 96\, x^2$ **Ans.** $C_1 + C_2 x + C_3\cos 2x + C_4\sin 2x + 2x^2(x^2 - 3)$

JECRC UNIVERSITY
BUILD YOUR WORLD

$$\boxed{\frac{1}{f(D)}.e^{ax}\cdot\phi(x)=e^{ax}.\frac{1}{f(D+a)}.\phi(x)}$$

$$D[e^{ax}\phi(x)]=e^{ax}D\phi(x)+ae^{ax}\phi(x)=e^{ax}(D+a)\phi(x)$$

$$D^2[e^{ax}\phi(x)]=D[e^{ax}(D+a)\phi(x)]=e^{ax}(D^2+aD)\phi(x)+ae^{ax}(D+a)\phi(x)$$

$$=e^{ax}(D^2+2aD+a^2)\phi(x)=e^{ax}(D+a)^2\phi(x)$$

Similarly, $D^n[e^{ax}\phi(x)]=e^{ax}(D+a)^n\phi(x)$

$$f(D)[e^{ax}\phi(x)]=e^{ax}f(D+a)\phi(x)$$

$$e^{ax}\phi(x)=\frac{1}{f(D)}.[e^{ax}f(D+a)\phi(x)] \quad ...(1)$$

Put $f(D+a)\phi(x)=X$, so that $\phi(x)=\frac{1}{f(D+a)}.X$

Substituting these values in (1), we get

$$e^{ax}\frac{1}{f(D+a)}X=\frac{1}{f(D)}[e^{ax}.X] \Rightarrow \frac{1}{f(D)}[e^{ax}.\phi(x)]=e^{ax}\frac{1}{f(D+a)}\phi(x)$$

Example 14. *Obtain the general solution of the differential equation*

$$y'' - 2y' + 2y = x + e^x \cos x. \qquad \textit{(U.P. II Semester Summer, 2002)}$$

Solution. $$y'' - 2y' + 2y = x + e^x \cos x$$

A.E. is $m^2 - 2m + 2 = 0 \quad \Rightarrow m = 1 \pm i$

$$\text{C.F.} = e^x (A \cos x + B \sin x)$$

$$\text{P.I.} = \frac{1}{D^2 - 2D + 2} x + \frac{1}{D^2 - 2D + 2} e^x \cos x$$

where $$I_1 = \frac{1}{D^2 - 2D + 2} x = \frac{1}{2\left[1 - D + \frac{D^2}{2}\right]} x = \frac{1}{2\left[1 - \left(D - \frac{D^2}{2}\right)\right]} x$$

$$= \frac{1}{2}\left[1 - \left(D - \frac{D^2}{2}\right)\right]^{-1} x = \frac{1}{2}\left[1 + \left(D - \frac{D^2}{2}\right) + \ldots\right] x = \frac{1}{2}\left[x + Dx - \frac{D^2}{2} x + \ldots\right] = \frac{1}{2}[x + 1]$$

and, $$I_2 = \frac{1}{D^2 - 2D + 2} e^x \cos x = e^x \frac{1}{(D+1)^2 - 2(D+1) + 2} \cos x = e^x \frac{1}{D^2 + 1} \cos x = e^x . x \frac{1}{2D} \cos x$$

$$\left[\text{If } f(-a^2) = 0, \text{ then } \frac{1}{f(D^2)} \cos ax = x \frac{1}{f'(-a^2)} \cos ax\right]$$

$$= \frac{1}{2} x e^x \sin x$$

$$y = \text{C.F.} + \text{P.I.}$$

Example 15. *Solve* : $(D^2 - 4D + 4)\, y = x^3 e^{2x}$

Solution. $(D^2 - 4D + 4)\, y = x^3 e^{2x}$

A.E. is $m^2 - 4m + 4 = 0 \quad \Rightarrow \quad (m-2)^2 = 0 \quad \Rightarrow \quad m = 2, 2$

C.F. $= (C_1 + C_2 x)\, e^{2x}$

$$\text{P.I.} = \frac{1}{D^2 - 4D + 4} x^3 \cdot e^{2x} = e^{2x} \frac{1}{(D+2)^2 - 4(D+2) + 4} x^3$$

$$= e^{2x} \frac{1}{D^2} x^3 = e^{2x} . \frac{1}{D}\left(\frac{x^4}{4}\right) = e^{2x} \cdot \frac{x^5}{20}$$

The complete solution is $y = (C_1 + C_2 x)\, e^{2x} + e^{2x} . \dfrac{x^5}{20}$ **Ans.**

Example 16. *Solve* :

$$\frac{d^4 y}{dx^4} - y = \cos x . \cosh x$$ (*Nagpur University, Summer* 2003)

Solution. We have, $(D^4 - 1)\, y = \cos x \cosh x$

A.E. is $m^4 - 1 = 0 \Rightarrow (m^2 - 1)(m^2 + 1) = 0 \quad \Rightarrow \quad m = \pm 1, \pm i$

C.F. $= C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x$

$$\text{P.I.} = \frac{1}{D^4 - 1} \cos x \;\; \cosh x = \frac{1}{D^4 - 1} \cos x \left(\frac{e^x + e^{-x}}{2}\right)$$

$$= \frac{1}{2}\left[\frac{1}{D^4 - 1} e^x \cos x + \frac{1}{D^4 - 1} e^{-x} \cos x\right] = \frac{1}{2}\left[e^x \frac{1}{(D+1)^4 - 1} \cos x + e^{-x} \frac{1}{(D-1)^4 - 1} \cos x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{D^4+4D^3+6D^2+4D}\cos x+e^{-x}\frac{1}{D^4-4D^3+6D^2-4D}\cos x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{(-1)^2+4D(-1)+6(-1)+4D}\cos x+e^{-x}\frac{1}{(-1)^2-4D(-1)+6(-1)-4D}\cos x\right]$$

$$= \frac{1}{2}\left[e^{x}\frac{1}{-5}\cos x+e^{-x}\frac{1}{-5}\cos x\right] = -\frac{1}{5}\left(\frac{e^{x}+e^{-x}}{2}\right)\cos x = -\frac{1}{5}\cosh x\cos x$$

Hence, the complete solution is

$$y = C_1e^{x}+C_2e^{-x}+C_3\cos x+C_4\sin x-\frac{1}{5}\cos x\cosh x$$ **Ans.**

Example 17. *Solve the differential equation :*

$$\frac{d^3y}{dx^3}-7\frac{d^2y}{dx^2}+10\frac{dy}{dx}=e^{2x}\sin x$$ (*AMIETE, June 2010, Nagpur University, Summer* 2005)

Solution. $$\frac{d^3y}{dx^3}-7\frac{d^2y}{dx^2}+10\frac{dy}{dx}=e^{2x}\sin x$$

$\Rightarrow$ $$D^3y - 7D^2y + 10\,Dy = e^{2x}\sin x$$

A.E. is

$$m^3 - 7m^2 + 10\,m = 0 \quad\Rightarrow\quad (m-2)(m^2-5m) = 0$$

$$\Rightarrow\quad m\,(m-2)\,(m-5) = 0 \quad\Rightarrow\quad m = 0, 2, 5$$

C.F $= C_1e^{0x}+C_2e^{2x}+C_3e^{5x}$

$$\text{P.I.} = \frac{1}{D^3 - 7D^2 + 10D} e^{2x} \sin x = e^{2x} \frac{1}{(D+2)^3 - 7(D+2)^2 + 10(D+2)} . \sin x$$

$$= e^{2x} \frac{1}{D^3 + 6D^2 + 12D + 8 - 7D^2 - 28D - 28 + 10D + 20} . \sin x$$

$$= e^{2x} \frac{1}{D^3 - D^2 - 6D} \sin x = e^{2x} \frac{1}{(-1^2) D - (-1^2) - 6D} \sin x$$

$$= e^{2x} \frac{1}{-D + 1 - 6D} \sin x = e^{2x} \frac{1}{1 - 7D} \sin x = e^{2x} \frac{1+7D}{1 - 49D^2} \sin x = e^{2x} \frac{1+7D}{1 - 49(-1^2)} \sin x$$

$$= e^{2x} \frac{1+7D}{50} \sin x = \frac{e^{2x}}{50} (\sin x + 7\cos x)$$

Complete solution is

$$y = \text{C.F.} + \text{P.I.}$$

$$\Rightarrow \qquad y = C_1 + C_2 e^{2x} + C_3 e^{5x} + \frac{e^{2x}}{50} (\sin x + 7\cos x)$$ **Ans.**

EXERCISE

JECRC UNIVERSITY
BUILD YOUR WORLD

Solve the following equations :

1. $(D^2 - 5D + 6)\,y = e^x \sin x$ **Ans.** $y = C_1 e^{2x} + C_2 e^{3x} + \frac{e^x}{10}(3\cos x + \sin x)$

2. $\frac{d^2 y}{dx^2} - 7\frac{dy}{dx} + 10y = e^{2x} \sin x$ **Ans.** $y = C_1 e^{2x} + C_2 e^{5x} + \frac{e^{2x}}{10}(3\cos x - \sin x)$

3. $\frac{d^3 y}{dx^3} - 2\frac{dy}{dx} + 4y = e^x \cos x$ **Ans.** $y = C_1 e^{-2x} + e^x(C_2 \cos x + C_3 \sin x) + \frac{xe^x}{20}(3\sin x - \cos x)$

4. $(D^2 - 4D + 3)\,y = 2xe^{3x} + 3e^{3x}\cos 2x$

Ans. $y = C_1 e^x + C_2 e^{3x} + \frac{1}{2}e^{3x}(x^2 - x) + \frac{3}{8}e^{3x}(\sin 2x - \cos 2x)$

JECRC Foundation

TO FIND THE VALUE OF $\dfrac{1}{f(D)}x^n \sin ax$.

Now $\dfrac{1}{f(D)}x^n(\cos ax + i\sin ax) = \dfrac{1}{f(D)}x^n e^{iax} = e^{iax}\dfrac{1}{f(D+ia)}x^n$

$$\boxed{\frac{1}{f(D)}\cdot x^n \sin ax = \text{Imaginary part of } e^{iax}\frac{1}{f(D+ia)}\cdot x^n}$$

$$\boxed{\frac{1}{f(D)}\cdot x^n \cos ax = \text{Real part of } e^{iax}\frac{1}{f(D+ia)}\cdot x^n}$$

Example 25. *Solve* $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + y = x\sin x$

Solution. Auxiliary equation is $m^2 - 2m + 1 = 0$ or $m = 1, 1$

C.F. $= (C_1 + C_2 x)e^x$

P.I. $= \dfrac{1}{D^2-2D+1}x\cdot\sin x \quad (e^{ix} = \cos x + i\sin x)$

$= \text{Imaginary part of } \dfrac{1}{D^2-2D+1}x(\cos x + i\sin x) = \text{Imaginary part of } \dfrac{1}{D^2-2D+1}x\cdot e^{ix}$

$= \text{Imaginary part of } e^{ix}\dfrac{1}{(D+i)^2-2(D+i)+1}\cdot x$

$= \text{Imaginary part of } e^{ix}\dfrac{1}{D^2-2(1-i)D-2i}\cdot x$

$= \text{Imaginary part of } e^{ix}\dfrac{1}{-2i}\left[1-(1+i)D-\dfrac{1}{2i}D^2\right]^{-1}\cdot x$

$$= \text{Imaginary part of } (\cos x + i\sin x)\left(\frac{i}{2}\right)[1+(1+i)D]x$$

$$= \text{Imaginary part of } \frac{1}{2}(i\cos x - \sin x)\,[x+1+i]$$

$$\text{P.I.} = \frac{1}{2}x\cos x + \frac{1}{2}\cos x - \frac{1}{2}\sin x$$

Complete solution is $y = (C_1 + C_2 x)e^x + \frac{1}{2}(x\cos x + \cos x - \sin x)$ **Ans.**

II- If $f(x)$ is in the form of $x \cdot V$, where V is any function of x:

$$\frac{1}{f(D)}x \cdot V = x\frac{1}{f(D)}V - \frac{f'(D)}{[f(D)]^2}V \quad \text{or} \quad x\frac{1}{f(D)}V + \frac{d}{dD}\left[\frac{1}{f(D)}\right]V$$

JECRC UNIVERSITY
BUILD YOUR WORLD

Q-2. Solve $(D^2+2D+1)y = x\sin x$.

Solve. A.E. $\Rightarrow m^2+2m+1=0 \Rightarrow (m+1)^2=0 \Rightarrow m=-1, -1$

$$C.F. = (C_1+C_2x)e^{-x}$$

$$P.I. = \frac{1}{(D^2+2D+1)} x\sin x$$

$$\therefore \frac{1}{f(D)} x.v = x\frac{1}{f(D)}v - \frac{f'(D)}{\{f(D)\}^2}v$$

$$\text{then} = x\frac{1}{(D^2+2D+1)}\sin x - \frac{2D+2}{(D^2+2D+1)^2}\sin x$$

$$= x\frac{1}{(\cancel{-1}+2D+\cancel{1})}\sin x - \frac{(2D+2)}{(\cancel{-1}+2D+\cancel{1})^2}\sin x$$

JECRC Foundation

JECRC UNIVERSITY
BUILD YOUR WORLD

$$= \frac{x}{2}\frac{1}{D}\sin x - \frac{2(D+1)}{4D^2}\sin x$$

$$= \frac{x}{2}(-\cos x) - \frac{1}{2}\frac{(D+1)}{(-1)}\sin x$$

$$= -\frac{x}{2}\cos x + \frac{1}{2}(D\sin x + \sin x)$$

$$= -\frac{x}{2}\cos x + \frac{1}{2}\cos x + \frac{1}{2}\sin x$$

Now complete solution is

$$y = C.F. + P.I. = (C_1 + C_2 x)e^{-x} - \frac{x}{2}\cos x + \frac{1}{2}(\cos x + \sin x)$$

JECRC Foundation

EXERCISE

Solve the following differential equations :

1. $(D^2+4)y = 3x\sin x$ **Ans.** $C_1\cos 2x + C_2\sin 2x + x\sin x - \frac{2}{3}\cos x$

2. $\frac{d^2y}{dx^2} - y = x\sin 3x + \cos x$ **Ans.** $C_1e^x + C_2e^{-x} - \frac{1}{10}\left[\frac{3}{5}\cos 3x + x\sin 3x + 5\cos x\right]$

3. $\frac{d^2y}{dx^2} - y = x\sin x + e^x + x^2e^x$ **Ans.** $C_1e^x + C_2e^{-x} - \frac{1}{2}[x\sin x + \cos x] + \frac{x}{12}e^x(2x^2 - 3x + 9)$

4. $(D^4 + 2D^2 + 1)\,y = x^2\cos x$

Ans. $(C_1 + C_2x)\cos x + (C_3 + C_4x)\sin x + \frac{1}{12}x^3\sin x - \frac{1}{48}(x^4 - 9x^2)\cos x$

GENERAL METHOD OF FINDING THE PARTICULAR INTEGRAL OF ANY FUNCTION ϕ (x)

$$\text{P.I.} = \frac{1}{D-a}\phi(x) = y \qquad \text{...(1)}$$

or $$(D-a)\frac{1}{D-a}\cdot\phi(x) = (D-a)\cdot y$$

$$\phi(x) = (D-a)y \qquad \text{or} \qquad \phi(x) = Dy - ay$$

$$\frac{dy}{dx} - ay = \phi(x)$$ which is the linear differential equation.

Its solution is $$ye^{-\int a\,dx} = \int e^{-\int a\,dx}\cdot\phi(x)\,dx \qquad \text{or} \qquad ye^{-ax} = \int e^{-ax}\cdot\phi(x)\,dx$$

$$y = e^{ax}\int e^{-ax}\cdot\phi(x)\,dx \qquad \boxed{\frac{1}{D-a}\cdot\phi(x) = e^{ax}\int e^{-ax}\cdot\phi(x)\,dx}$$

Example 26. *Solve* $\dfrac{d^2y}{dx^2}+9y=\sec 3x.$

Solution. Auxiliary equation is $m^2+9=0$ or $m=\pm 3i$,

$$\text{C.F.} = C_1 \cos 3x + C_2 \sin 3x$$

$$\text{P.I.} = \frac{1}{D^2+9}\cdot\sec 3x = \frac{1}{(D+3i)\,(D-3i)}\cdot\sec 3x = \frac{1}{6i}\left[\frac{1}{D-3i}-\frac{1}{D+3i}\right]\cdot\sec 3x$$

$$= \frac{1}{6i}\cdot\frac{1}{D-3i}\cdot\sec 3x - \frac{1}{6i}\frac{1}{D+3i}\cdot\sec 3x \qquad \ldots(1)$$

Now, $\dfrac{1}{D-3i}\sec 3x = e^{3ix}\displaystyle\int e^{-3ix}\sec 3x\,dx$ $\qquad \left[\dfrac{1}{D-a}\phi(x) = e^{ax}\displaystyle\int e^{-ax}\phi(x)\,dx\right]$

$$= e^{3ix}\int\frac{\cos 3x - i\sin 3x}{\cos 3x}dx = e^{3ix}\int(1-i\tan 3x)\,dx = e^{3ix}(x+\frac{i}{3}\log\cos 3x)$$

Changing i to $-i$, we have $\dfrac{1}{D+3i}\sec 3x = e^{-3ix}(x-\dfrac{i}{3}\log\cos 3x)$

Putting these values in (1), we get

$$\text{P.I.} = \frac{1}{6i}\left[e^{3ix}\left(x+\frac{i}{3}\log\cos 3x\right)-e^{-3ix}\left(x-\frac{i}{3}\log\cos 3x\right)\right]$$

$$= \frac{x}{6i}e^{3ix}+\frac{e^{3ix}\log\cos 3x}{18}-\frac{xe^{-3ix}}{6i}+\frac{e^{-3ix}}{18}\log\cos 3x$$

$$= \frac{x}{3}\frac{e^{3ix}-e^{-3ix}}{2i}+\frac{1}{9}\cdot\frac{e^{3ix}+e^{-3ix}}{2}\log\cos 3x = \frac{x}{3}\sin 3x+\frac{1}{9}\cos 3x\cdot\log\cos 3x$$

Hence, complete solution is $y = C_1\cos 3x + C_2\sin 3x + \frac{x}{3}\sin 3x + \frac{1}{9}\cos 3x\cdot\log\cos 3x$ **Ans.**

Example 27. *Solve* $\frac{d^2y}{dx^2}+2\frac{dy}{dx}+2y = e^{-x}\sec^3 x.$ *(Nagpur, Winter 2000)*

Solution. Here we have

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}+2y = e^{-x}\sec^3 x$$

$$(D^2+2D+2)y = e^{-x}\sec^3 x$$

A.E. is $m^2+2m+2=0$ $\Rightarrow m = \frac{-2\pm\sqrt{4-8}}{2} = -1+i$

$C.F. = e^{-x}(C_1\cos x + C_2\sin x)$

$$\text{P.I.} = \frac{1}{D^2+2D+2}e^{-x}\sec^3 x = e^{-x}\frac{1}{(D-1)^2+2(D-1)+2}\sec^{3x}$$

$$= e^{-x}\frac{1}{D^2+1}\sec^{3x} = \frac{1}{(D+i)(D-i)}\sec^3 x = e^{-x}\frac{1}{2i}\left[\frac{1}{D-i}-\frac{1}{D+i}\right]\sec^3 x \quad \ldots(1)$$

Now $\frac{1}{D-i}\sec^3 x = e^{ix}\int e^{-ix}\sec^3 x\,dx$ $\left[\frac{1}{D-a}\phi(x) = e^{ax}\int e^{-ax}\phi(x)\,dx\right]$

$$= e^{ix}\int(\cos x - i\sin x)\sec^3 x\,dx = e^{ix}\int[\sec^2 x - i\tan x\sec^2 x]dx = e^{ix}\left[\tan x - \frac{i\tan^2 x}{2}\right] \ldots(2)$$

Similarly $\frac{1}{D+i}\sec^3 x = e^{-ix}\left[\tan x + i\frac{\tan^2 x}{2}\right]$ ...(3) [changing i to $-i$]

Putting the values from (2) and (3) in (1), we get

$$P.I. = \frac{e^{-x}}{2i}\left[e^{ix}\left(\tan x - \frac{i\tan^2 x}{2}\right) - e^{-ix}\left(\tan x + i\frac{\tan^2 x}{2}\right)\right]$$

$$= e^{-x}\left[\tan x\frac{e^{ix}-e^{-ix}}{2i} - i\frac{\tan^2 x}{2i}\left(\frac{e^{ix}+e^{-ix}}{2}\right)\right] = e^{-x}\left[\tan x\sin x - \frac{\tan^2 x}{2}\cos x\right]$$

$$= e^{-x}\left[\tan x.\sin x - \frac{\tan x}{2}\frac{\sin x}{\cos x}\cos x\right] = e^{-x}\left[\tan x\sin x - \frac{\tan x}{2}\sin x\right] = e^{-x}\left(\frac{1}{2}\tan x\sin x\right)$$

Complete solution = C.F. + P.I.

$$= e^{-x}(c_1\cos x + c_2\sin x) + \frac{e^{-x}}{2}\tan x\sin x = e^{-x}\left[C_1\cos x + C_2\sin x + \frac{\sin x\tan x}{2}\right]$$ **Ans.**

Example 28. *Solve* $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = e^x\tan x$.

Solution. Here we have $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = e^x\tan x$

$(D^2 - 2D + 2)y = e^x\tan x$

A.E. is $m^2 - 2m + 2 = 0$ $\Rightarrow m = \frac{2\pm\sqrt{4-8}}{2} = 1\pm i$

$C.F. = e^x(C_1\cos x + C_2\sin x)$

$$P.I. = \frac{1}{D^2 - 2D + 2} e^{x} \tan x = e^x \frac{1}{(D+1)^2 - 2(D+1) + 2} \tan x$$

$$= e^x \frac{1}{D^2 + 1} \tan x = e^x \frac{1}{(D+i)(D-i)} \tan x = \frac{e^x}{2i}\left[\frac{1}{D-i} - \frac{1}{D+i}\right] \tan x \qquad \ldots(1)$$

Now $\frac{1}{D-i} \tan x = e^{ix} \int e^{-ix} \tan x \, dx$

$$= e^{ix} \int (\cos x - i \sin x) \tan x \, dx = e^{ix} \int \left(\sin x - i \frac{\sin^2 x}{\cos x} \right) dx$$

$$= e^{ix} \int \left[\sin x - \frac{i(1 - \cos^2 x)}{\cos x} \right] dx = e^{ix} \int (\sin x - i \sec x + i \cos x] dx$$

$$= e^{ix} [-\cos x + i \sin x - i \log(\sec x + \tan x)] \qquad \ldots(2)$$

Similarly $\frac{1}{D+i} \tan x = e^{-ix} [-\cos x - i \sin x + i \log(\sec x + \tan x)] \qquad \ldots(3)$

On putting the values from (2) and (3) in (1), we get

$$P.I. = \frac{e^x}{2i} [e^{ix}(-\cos x + i \sin x - i \log(\sec x + \tan x) - e^{-ix}(-\cos x - i \sin x + i \log(\sec x + \tan x)]$$

$$= e^x \left[-\cos x \frac{e^{ix} - e^{-ix}}{2i} + \sin x \frac{e^{ix} + e^{-ix}}{2} - \log(\sec x + \tan x) \frac{e^{ix} + e^{-ix}}{2} \right]$$

$$= e^x [-\cos x \sin x + \sin x \cos x - \cos x \log(\sec x + \tan x)]$$

$$= -e^x \cos x \log(\sec x + \tan x)$$

Complete solution = C.F. + P.I.

$$= e^x (C_1 \cos x + C_2 \sin x) - e^x \cos x \log(\sec x + \tan x) \qquad \textbf{Ans.}$$

EXERCISE

Solve the following differential equations :

1. $\dfrac{d^2y}{dx^2}+a^2y=\sec ax$ *(R.G.P.V., Bhopal April, 2010)*

Ans. $C_1\cos ax+C_2\sin ax+\dfrac{x}{a}\sin ax+\dfrac{1}{a^2}\cos ax\cdot\log\cos ax$

2. $\dfrac{d^2y}{dx^2}+y=\operatorname{cosec} x$ **Ans.** $C_1\cos x+C_2\sin x-x\cos x+\sin x\log\sin x$

3. $(D^2+4)\,y=\tan 2x$ **Ans.** $C_1\cos 2x+C_2\sin 2x-\dfrac{1}{4}\cos 2x\log(\sec 2x+\tan 2x)$

4. $\dfrac{d^2y}{dx^2}+y=(x-\cot x)$ *(A.M.I.E. Winter 2002)*

Ans. $C_1\cos x+C_2\sin x-x\cos^2 x-\sin x\log(\operatorname{cosec} x-\cot x)$

Second Order Linear Differential Equation with Variable Coefficients

Second order linear differential equation with variable coefficients :

A differential equation of the type

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R \quad \text{———} \quad (1)$$

where P, Q, R are functions of x, is know as second order linear diff. equation with variable coefficients. the method of solution of (1) depends upon the nature of functions P(x), Q(x) and R(x).

$$a_n x_n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{d x^{n-1}} + \dots + a_0 y = \phi(x) \qquad \dots (1)$$

where a_0, a_1, a_2, ... are constants, is called a homogeneous equation.

Put $\quad x = e^z, \quad z = \log_e x, \quad \frac{d}{dz} \equiv D$

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{1}{x} \frac{dy}{dz} \Rightarrow x \frac{dy}{dx} = \frac{dy}{dz} \Rightarrow x \frac{dy}{dx} = Dy$$

Again,
$$\frac{d^2 y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d^2 y}{dz^2}\frac{dz}{dx}$$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d^2 y}{dz^2}\frac{1}{x} = \frac{1}{x^2}\left(\frac{d^2 y}{dz^2} - \frac{dy}{dz}\right) = \frac{1}{x^2}(D^2 - D)\, y$$

$$x^2 \frac{d^2 y}{dx^2} = (D^2 - D)\, y$$

or
$$\boxed{x^2 \frac{d^2 y}{dx^2} = D(D-1)\, y}$$

Similarly.
$$\boxed{x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)\, y}$$

The substitution of these values in (1) reduces the given homogeneous equation to a differential equation with constant coefficients.

Example 1. *Solve*: $$x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$$ (*A.M.I.E. Summer* 2000)

Solution. We have, $$x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4 \qquad \dots (1)$$

Putting $x = e^z$, $D \equiv \frac{d}{dz}$, $x \frac{dy}{dx} = Dy$, $x^2 \frac{d^2 y}{dx^2} = D(D-1)\, y$ in (1), we get

$$D(D-1)\, y - 2\, Dy - 4y = e^{4z} \quad \text{or} \quad (D^2 - 3D - 4)\, y = e^{4z}$$

A.E. is $m^2 - 3m - 4 = 0 \Rightarrow (m-4)(m+1) = 0 \Rightarrow m = -1, 4$

$$\text{C.F.} = C_1 e^{-z} + C_2 e^{4z}$$

$$\text{P.I.} = \frac{1}{D^2 - 3D - 4} e^{4z} \qquad \text{[Rule Fails]}$$

$$= z \frac{1}{2D-3} e^{4z} = z \frac{1}{2(4) - 3} e^{4z} = \frac{z e^{4z}}{5}$$

Thus, the complete solution is given by

$$y = C_1 e^{-z} + C_2 e^{4z} + \frac{z e^{4z}}{5} \Rightarrow y = \frac{C_1}{x} + C_2 x^4 + \frac{1}{5} x^4 \log x \quad \textbf{Ans.}$$

Example 2. *Solve:* $x^2\dfrac{d^2y}{dx^2}+4x\dfrac{dy}{dx}+2y=e^x$

(U.P., II Semester 2005; Nagpur University, Summer 2001)

Solution. Given equation is $x^2\dfrac{d^2y}{dx^2}+4x\dfrac{dy}{dx}+2y=e^x$ ...(1)

To solve (1), we put $x = e^z$ or $z = \log x$ and $D\equiv\dfrac{d}{dz}$

$$x\frac{dy}{dx}=Dy \quad \text{and} \quad x^2\frac{d^2y}{dx^2}=D(D-1)\,y$$

Substituting these values in (1), it reduces

$$[\,D\,(D-1)+4\,D+2\,]\,y = e^{e^z} \quad \Rightarrow \quad (D^2+3D+2)\,y=e^{e^z}$$

$\therefore$ It's $A.E.$ is $m^2+3m+2=0$

$\therefore$ $(m+1)(m+2)=0$ $\therefore$ $m=-1,-2$

$\therefore$ $$\text{C.F.} = C_1e^{-z}+C_2e^{-2z}=\frac{C_1}{x}+\frac{C_2}{x^2}$$

$$\text{P.I.} = \frac{1}{D^2+3D+2}e^{e^z}=\frac{1}{(D+1)(D+2)}e^{e^z}$$

$$=\left(\frac{1}{D+1}-\frac{1}{D+2}\right)e^{e^z}=\frac{1}{D+1}e^{e^z}-\frac{1}{D+2}e^{e^z}$$

$$= e^{-z}\int e^{e^z}\,.\,e^z\,dz - e^{-2z}\int e^{e^z}\,.\,e^{2z}\,dz \qquad \left[\text{Since } \frac{1}{D-a}X = e^{ax}\int Xe^{-ax}\,dx\right]$$

Put $\quad e^z = t$ so that $e^z dz = dt$

$$\text{P.I.} = e^{-z}\int e^t\,dt - e^{-2z}\int e^t\,.\,t\,dt = e^{-z}e^t - e^{-2z}\,(te^t - e^t)$$

$$= e^{-z}e^{e^z} - e^{-2z}\,(e^z\,e^{e^z} - e^{e^z})$$

$$= e^{-z}e^{e^z} - e^{-z}e^{e^z} + e^{-2z}\,e^{e^z} = e^{-2z}\,e^{e^z} = x^{-2}e^x = \frac{e^x}{x^2} \qquad (\because\ x = e^z)$$

Hence, the C.S. of (1) is $\quad y = \text{C.F.} + \text{P.I.}$

$$y = \frac{C_1}{x} + \frac{C_2}{x^2} + \frac{e^x}{x^2}$$

Ans.

Example 3. *Solve* $(x^3 D^3 + x^2 D^2 - 2)\, y = x - \dfrac{1}{x^3}$ *(Nagpur University, Summer 2000)*

Solution. Put $x = e^z$, $z = \log x$

Let $D_1 \equiv \dfrac{d}{dz}$

Then $x^2 \dfrac{d^2 y}{dx^2} = D_1 (D_1 - 1)\, y \quad \Rightarrow \quad x^3 \dfrac{d^3 y}{dx^3} = D_1 (D_1 - 1)(D_1 - 2)\, y$

Substituting in the given equation, we get

$$[\, D_1 (D_1 - 1)(D_1 - 2) + D_1 (D_1 - 1) - 2 \,]\, y = e^z + e^{-3z}$$

$$\Rightarrow \quad [\, D_1^3 - 3D_1^2 + 2D_1 + D_1^2 - D_1 - 2 \,]\, y = e^z + e^{-3z}$$

$$\Rightarrow \quad (\, D_1^3 - 2D_1^2 + D_1 - 2 \,)\, y = e^z + e^{-3z}$$

A.E. is $m^3 - 2m^2 + m - 2 = 0$

i.e. $(m - 2)(m^2 + 1) = 0$ *i.e.* $m = 2, \ \pm i.$

$$\text{C.F.} = C_1 e^{2z} + C_2 \cos z + C_3 \sin z$$

$$\text{P.I.} = \frac{1}{D_1^3 - 2D_1^2 + D_1 - 2} e^z + \frac{1}{D_1^3 - 2D_1^2 + D_1 - 2} e^{-3z}$$

$$= \frac{1}{1 - 2 + 1 - 2} e^z + \frac{1}{-27 - 18 - 3 - 2} e^{-3z} \qquad \begin{bmatrix} D_1 = 1 \\ D_1 = -3 \end{bmatrix}$$

$$= -\frac{e^z}{2} - \frac{1}{50} e^{-3z}$$

$\therefore$ $$y = C_1 e^{2z} + C_2 \cos z + C_3 \sin z - \frac{1}{2} e^{z} - \frac{1}{50} e^{-3z}$$

$$= C_1 x^2 + C_2 \cos(\log x) + C_3 \sin(\log x) - \frac{1}{2} x - \frac{1}{50} \cdot \frac{1}{x^3}$$ **Ans.**

Example 4. *Solve* $x^2 \dfrac{d^2 y}{dx^2} + x \dfrac{dy}{dx} + y = \sin(\log x^2)$ (*Nagpur University, Summer* 2005)

Solution. We have, $$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \sin(\log x^2) \quad \ldots (1)$$

Let $x = e^z$, so that $z = \log x$, $D \equiv \dfrac{d}{dz}$

(1) becomes

$$D(D-1)y + Dy + y = \sin(2z) \quad \Rightarrow \quad (D^2 + 1)y = \sin 2z$$

A.E. is $m^2 + 1 = 0$ or $m = \pm i$

$C.F. = C_1 \cos z + C_2 \sin z$

$$\text{P.I} = \frac{1}{D^2 + 1} \sin 2z = \frac{1}{-4+1} \sin 2z = -\frac{1}{3} \sin 2z$$

$$y = \text{C.F.} + \text{P.I.} = C_1 \cos z + C_2 \sin z - \frac{1}{3} \sin 2z$$

$$= C_1 \cos(\log x) + C_2 \sin(\log x) - \frac{1}{3} \sin(\log x^2)$$ **Ans.**

Example 6. *Solve:* $x^2 \dfrac{d^3y}{dx^3} + 3x \dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = x^2 \log x$ (*Nagpur University, Summer* 2003)

Solution. We have, $x^3 \dfrac{d^3y}{dx^3} + 3x^2 \dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} = x^3 \log x$

Let $x = e^z$ so that $z = \log x$, $D \equiv \dfrac{d}{dz}$

The equation becomes after substitution

$$[D(D-1)(D-2) + 3D(D-1) + D]\, y = z\, e^{3z} \quad \Rightarrow \quad D^3 y = z e^{3z}$$

Auxiliary equation is $m^3 = 0 \Rightarrow \quad m = 0, 0, 0.$

C.F. $= C_1 + C_2 z + C_3 z^2 = C_1 + C_2 \log x + C_3 (\log x)^2$

$$\text{P.I.} = \frac{1}{D^3} \cdot z e^{3z} = e^{3z} \cdot \frac{1}{(D+3)^3} \cdot z$$

$$= e^{3z} \frac{1}{27}\left(1+\frac{D}{3}\right)^{-3} z = \frac{e^{3z}}{27}(1-D)\, z = \frac{e^{3z}}{27}(z-1) = \frac{x^3}{27}(\log x - 1)$$

Complete solution is $\quad y = C_1 + C_2 \log x + C_3 (\log x)^2 + \dfrac{x^3}{27}(\log x - 1)$ **Ans.**

Example 7. *Solve the differential equation*

$$x^2 \frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x \log x.$$ (*Nagpur University, Winter 2002, Summer 2000*)

Solution. Putting $x = e^z$ or $z = \log x$ and $D \equiv \frac{d}{dz}$

On putting $x\frac{dy}{dx} = Dy$ and $x^2\frac{d^2y}{dx^2} = D(D-1)y$ in the given equation, we get

$$[D(D-1) - 3D + 5]\, y = ze^z$$

i.e. $$(D^2 - 4D + 5)\, y = ze^z$$

It's A.E. is $$m^2 - 4m + 5 = 0$$

$$\therefore \quad m = \frac{4 \pm \sqrt{16-20}}{2} = \frac{4 \pm 2i}{2} = 2 \pm i$$

$$\text{C.F.} = e^{2z}(C_1 \cos z + C_2 \sin z)$$

$$\text{P.I.} = \frac{1}{D^2 - 4D + 5} ze^z$$

$$= e^z \cdot \frac{1}{(D+1)^2 - 4(D+1) + 5} z$$ (by replacing D by $D + 1$)

$$= e^z \cdot \frac{1}{D^2 - 2D + 2} z = \frac{e^z}{2} \cdot \frac{1}{1 + \left(\frac{D^2 - 2D}{2}\right)} z$$

$$= \frac{e^z}{2}\left[1+\left(\frac{D^2-2D}{2}\right)\right]^{-1} z = \frac{e^z}{2}\left[1-\left(\frac{D^2-2D}{2}\right)+\ldots\right] z$$

$$= \frac{e^z}{2}[z + Dz] = \frac{e^z}{2}(z+1).$$

Hence, the solution is

$$y = e^{2z}[C_1 \cos z + C_2 \sin z] + \frac{e^z}{2}(z+1)$$

$$y = x^2 [C_1 \cos(\log x) + C_2 \sin(\log x)] + \frac{x}{2}(1+\log x)$$ **Ans.**

Example 8. *Solve the homogeneous linear differential equation.*

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = (\log x)\sin(\log x)$$

(*Nagpur University, Winter* 2002, *U.P. II Semester, Summer* 2002)

Solution. Since given equation is homogeneous,

Put $x = e^z \quad \Rightarrow \log x = z$

Also, $x\frac{dy}{dx} = Dy, \quad x^2\frac{d^2y}{dx^2} = D(D-1)y, \quad D \equiv \frac{d}{dz}$

The transformed equation is

$$D(D-1)y + Dy + y = z \sin z$$
$$(D^2 - D + D + 1)y = z\sin z$$
$$(D^2+1)y = z \sin z$$

A.E. is $$m^2 + 1 = 0 \quad \Rightarrow \quad m = \pm i$$

$$\text{C.F.} = C_1 \cos z + C_2 \sin z$$

$$P.I. = \frac{1}{D^2+1} z \sin z$$

$$= \text{Imaginary part of } \frac{1}{D^2+1} z(\cos z + i \sin z)$$

$$= \text{Imaginary part of } \frac{1}{D^2+1} ze^{iz} \qquad = \text{Imaginary part of } e^{iz} \frac{1}{(D+i)^2+1} z$$

$$= \text{Imaginary part of } e^{iz} \frac{1}{D^2+2iD-1+1} z \quad = \text{Imaginary part of } e^{iz} \frac{1}{D^2+2iD} z$$

$$= \text{Imaginary part of } e^{iz} \frac{1}{2iD} \frac{1}{\left(1+\frac{D}{2i}\right)} z \qquad = \text{Imaginary part of } e^{iz} \frac{1}{2iD} \left(1-\frac{D}{2i}\right) z$$

$$= \text{Imaginary part of } e^{iz} \frac{1}{2iD} \left(z-\frac{1}{2i}\right) \qquad = \text{Imaginary part of } e^{iz} \frac{1}{2i} \left(\frac{z^2}{2}-\frac{z}{2i}\right)$$

$$= \text{Imaginary part of } \frac{1}{2i} (\cos z + i \sin z) \left(\frac{z^2}{2}-\frac{z}{2i}\right)$$

$$= \text{Imaginary part of } (\cos z + i \sin z) \left(\frac{z^2}{4i}+\frac{z}{4}\right)$$

$= \text{Imaginary part of } (\cos z + i\sin z)\left(-i\,\frac{z^2}{4} + \frac{z}{4}\right) = -\frac{z^2}{4}\cos z + \frac{z}{4}\sin z$

Complete solution is, y = C.F. + P.I.

$$y = C_1 \cos z + C_2 \sin z - \frac{z^2}{4}\cos z + \frac{z}{4}\sin z$$

$$y = C_1 \cos(\log x) + C_2 \sin(\log x) - \frac{1}{4}(\log x)^2 \cos(\log x) + \frac{1}{4}(\log x)\sin(\log x)$$ **Ans.**

EXERCISE

Solve the following differential equations:

1. $x^2 \dfrac{d^2y}{dx^2} - 4x\dfrac{dy}{dx} + 6y = \dfrac{42}{x^4}$ **Ans.** $C_1 x^2 + C_2 x^3 + \dfrac{1}{x^4}$

2. $(x^2 D^2 - 3xD + 4)\, y = 2x^2$ **Ans.** $(C_1 + C_2 \log x)\, x^2 + x^2 (\log x)^2$

3. $x^2 \dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + y = \log x$ *(AMIETE, June 2010)* **Ans.** $(C_1 + C_2 \log x)\, x + \log x + 2$

4. $\dfrac{d^2y}{dx^2} + \dfrac{1}{x}\dfrac{dy}{dx} = \dfrac{12 \log x}{x^2}$ **Ans.** $C_1 + C_2 \log x + 2(\log x)^3$

JECRC UNIVERSITY
BUILD YOUR WORLD

LEGENDRE'S HOMOGENEOUS DIFFERENTIAL EQUATIONS

A linear differential equation of the form

$$(a+bx)^n \frac{d^n y}{dx^n} + a_1 (a+bx)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \ldots + a_n y = X \qquad \ldots (1)$$

where a, b, a_1, a_2, ... a_n are constants and X is a function of x, is called Legendre's linear equation.

Equation (1) can be reduced to linear differential equation with constant coefficients by the substitution.

$$a + bx = e^z \qquad \Rightarrow \qquad z = \log (a + bx)$$

so that $$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{b}{a+bx} \cdot \frac{dy}{dz}$$

$$\Rightarrow \qquad (a+bx)\frac{dy}{dx} = b\frac{dy}{dz} = b\,Dy, \quad D \equiv \frac{d}{dz} \qquad \Rightarrow \qquad (a+bx)\frac{dy}{dx} = b\,Dy$$

where

Again $$\frac{d^2 y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{b}{a+bx} \cdot \frac{dy}{dz}\right)$$

$$= -\frac{b^2}{(a+bx)^2}\frac{dy}{dz} + \frac{b}{(a+bx)} \cdot \frac{d^2 y}{dz^2} \cdot \frac{dz}{dx}$$

$$= -\frac{b^2}{(a+bx)^2}\frac{dy}{dz} + \frac{b}{(a+bx)}\cdot\frac{d^2y}{dz^2}\cdot\frac{b}{(a+bx)}$$

$$\Rightarrow \quad (a+bx)^2\frac{d^2y}{dx^2} = -b^2\frac{dy}{dz} + b^2\frac{d^2y}{dz^2}$$

$$= b^2\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right) = b^2\,(D^2\,y - D\,y) = b^2 D\,(\,D-1)y$$

$$\Rightarrow \quad (a+bx)^2\frac{d^2y}{dx^2} = b^2\;D\,(D-1)$$

Similarly, $(a+bx)^3\dfrac{d^3y}{dx^3} = b^3\;D(D-1)\,(D-2)y$

....

$$(a+bx)^n\frac{d^ny}{dx^n} = b^n\;D\,(D-1)\,(D-2)\ldots\ldots(D-n+1)y$$

Substituting these values in equation (1), we get a linear differential equation with constant coefficients, which can be solved by the method given in the previous section.

Example 1 . *Solve* $(1+x)^2 \dfrac{d^2y}{dx^2} + (1+x)\dfrac{dy}{dx} + y = \sin 2\{\log(1+x)\}$

Solution. We have, $(1+x)^2 \dfrac{d^2y}{dx^2} + (1+x)\dfrac{dy}{dx} + y = \sin 2\{\log(1+x)\}$

Put $1 + x = e^z$ or $\log(1+x) = z$

$(1+x)\dfrac{dy}{dx} = Dy$ and $(1+x)^2\dfrac{d^2y}{dx^2} = D(D-1)y$, where $D \equiv \dfrac{d}{dz}$

Putting these values in the given differential equation, we get

$$D(D-1)y + Dy + y = \sin 2z \quad \text{or} \quad (D^2 - D + D + 1)y = \sin 2z$$

$$(D^2+1)y = \sin 2z$$

A.E. is $m^2 + 1 = 0 \quad \Rightarrow \quad m = \pm i$

$$C.F. = A\cos z + B\sin z$$

$$\text{P.I.} = \frac{1}{D^2+1}\sin 2z = \frac{1}{-4+1}\sin 2z = -\frac{1}{3}\sin 2z$$

Now, complete solution is y = C.F. + P.I.

$$\Rightarrow \quad y = A\cos z + B\sin z - \frac{1}{3}\sin 2z$$

$$\Rightarrow \quad y = A\cos\{\log(1+x)\} + B\sin\{\log(1+x)\} - \frac{1}{3}\sin 2\{\log(1+x)\}$$ **Ans.**

JECRC UNIVERSITY
BUILD YOUR WORLD

Example 2. *Solve:* $(3x+2)^2 \dfrac{d^2y}{dx^2} + 3\,(3x+2)\dfrac{dy}{dx} - 36\,y = 3x^2 + 4x + 1.$

(*Uttarakhand II, Semester, June* 2007)

Solution. Let $3x + 2 = e^z \quad \Rightarrow \quad z = \log(3x+2) \qquad \left[\, x = \dfrac{e^z - 2}{3} \,\right]$

So that $(3x+2)\dfrac{dy}{dx} = 3\,Dy$ and $(3x+2)^2 \dfrac{d^2y}{dx^2} = 9D\,(D-1)y$ where $D \equiv \dfrac{d}{dz}$

Putting these values in the given differential equation, we get

$$9D\,(D-1)\,y + 9Dy - 36y = 3\left(\frac{e^z - 2}{3}\right)^2 + 4\left(\frac{e^z - 2}{3}\right) + 1$$

$$\therefore \quad (9D^2 - 36)\,y = \frac{1}{3}(e^{2z} - 4\,e^z + 4) + \frac{4}{3}e^z - \frac{8}{3} + 1 = \frac{e^{2z}}{3} - \frac{1}{3}$$

A.E. is $\quad 9m^2 - 36 = 0$

$$m^2 - 4 = 0 \qquad \therefore \quad m = \pm 2$$

$$\text{C.F.} = C_1\,e^{2z} + C_2\,e^{-2z}$$

$$\text{P.I.} = \frac{1}{9D^2 - 36}\left[\frac{e^{2z}}{3} - \frac{1}{3}\right] = \frac{1}{27}\,\frac{1}{D^2 - 4}\,e^{2z} - \frac{1}{3}\,\frac{1}{9D^2 - 36}\,e^{0z}$$

$$= \frac{1}{27} z \frac{1}{2D} e^{2z} - \frac{1}{3} \frac{1}{0-36} = \frac{1}{27} z \left(\frac{e^{2z}}{4} \right) + \frac{1}{108} = \frac{1}{108} [ze^{2z} + 1]$$

Complete Solution is

$$y = C.F. + P.I. = C_1 e^{2z} + C_2 e^{-2z} + \frac{1}{108} (ze^{2z} + 1)$$

$$y = C_1 (3x+2)^2 + \frac{C_2}{(3x+2)^2} + \frac{1}{108} [(3x+2)^2 \log (3x+2) + 1]$$ **Ans.**

EXERCISE

1. $(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 2 \sin \log (1+x)$

Ans. $y = C_1 \cos \log (1+x) + C_2 \sin \log (1+x) - \log (1+x) \cos \log (1+x)$

METHOD OF VARIATION OF PARAMETERS

To find particular integral of

$$\frac{d^2 y}{dx^2} + b\frac{dy}{dx} + c\,y = X \qquad \text{... (1)}$$

Let complementary function = $Ay_1 + By_2$, so that y_1 and y_2 satisfy

$$\frac{d^2 y}{dx^2} + b\frac{dy}{dx} + c\,y = 0 \qquad \text{... (2)}$$

Let us assume particular integral $y = uy_1 + vy_2$, ... (3)

where u and v are unknown functions of x.

Differentiation (3) *w.r.t.* x, we have $y' = uy_1' + vy_2' + u'y_1 + v'y_2$ assuming that u,v satisfy the equation

$$u'y_1 + v'y_2 = 0 \qquad \text{... (4)}$$

then $$y' = uy_1' + vy_2' \qquad \text{... (5)}$$

Differentiating (5) *w.r.t.x*, we have $y'' = uy_1'' + u'y_1' + vy_2'' + v'y_2'$

Substituting the values of y, y' and y'' in (1), we get

$$(uy_1'' + u'y_1' + vy_2'' + v'y_2') + b\,(uy_1' + vy_2') + c\,(uy_1 + vy_2) = X$$

$\Rightarrow$ $$u(y_1'' + by_1' + cy_1) + v(y_2'' + by_2' + cy_2) + (u'y_1' + v'y_2') = X \quad \text{... (6)}$$

y_1 and y_2 will satisfy equation (1)

$\therefore$ $$y_1'' + by_1' + cy_1 = 0 \quad \text{... (7)}$$

and $$y_2'' + by_2' + cy_2 = 0 \quad \text{... (8)}$$

Putting the values of expressions from (7) and (8) in (6), we get

$\Rightarrow$ $$u'y_1' + v'y_2' = X \quad \text{... (9)}$$

Solving (4) and (9), we get

$$u' = \begin{vmatrix} 0 & y_2 \\ X & y_2' \end{vmatrix} \div \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \frac{-y_2 X}{y_1 y_2' - y_1' y_2}$$

$$v' = \begin{vmatrix} y_1 & 0 \\ y_1' & X \end{vmatrix} \div \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \frac{y_1 X}{y_1 y_2' - y_1' y_2}$$

$$\boxed{u = \int \frac{-y_2 X}{y_1 y_2' - y_1' y_2}\,dx} \qquad \boxed{v = \int \frac{y_1 X}{y_1 y_2' - y_1' y_2}\,dx}$$

Working Rule

Step 1. Find out the *C.F. i.e.*, $A\,y_1 + B\,y_2$

Step 2. Particular integral = $u\,y_1 + v\,y_2$

Step 3. Find u and v by the formulae

$$u = \int \frac{-y_2\,X}{y_1\,y_2' - y_1'\,y_2}\,dx, \quad v = \int \frac{y_1\,X}{y_1\,y_2' - y_1'\,y_2}\,dx$$

Example 1 *Solve* $\frac{d^2y}{dx^2} + y = \text{cosec}\, x.$

Solution. $(D^2 + 1)\, y = \text{cosec}\, x$

A.E. is $m^2 + 1 = 0 \quad \Rightarrow \quad m = \pm\, i$

$$\text{C.F.} = A \cos x + B \sin x$$

Here $y_1 = \cos x, \qquad y_2 = \sin x$

$$\text{P.I.} = y_1\, u + y_2\, v$$

where
$$u = \int \frac{-y_2 \,.\, \text{cosec}\, x\, dx}{y_1 \,.\, y'_2 - y'_1 \,.\, y_2} = \int \frac{-\sin x \,.\, \text{cosec}\, x\, dx}{\cos x\,(\cos x) - (-\sin x)\,(\sin x)}$$

$$= \int \frac{-\sin x \,.\, \frac{1}{\sin x}\, dx}{\cos^2 x + \sin^2 x} = -\int dx = -x$$

$$v = \int \frac{y_1 \,.\, X\, dx}{y_1 \,.\, y'_2 - y'_1\, y_2} = \int \frac{\cos x \,.\, \text{cosec}\, x\, dx}{\cos x\,(\cos x) - (-\sin x)\,(\sin x)}$$

$$= \int \frac{\cos x \,.\, \frac{1}{\sin x}}{\cos^2 x + \sin^2 x}\, dx = \int \frac{\cot x\, dx}{1} = \log \sin x$$

$$P.I. = uy_1 + vy_2 = -x \cos x + \sin x\, (\log \sin x)$$

General solution = C.F. + P.I.

$$y = A \cos x + B \sin x - x \cos x + \sin x\, .(\log \sin x)$$ **Ans.**

Example : *Apply the mehtod of variation of parameters to solve*

$$\frac{d^2y}{dx^2} + y = \tan x$$

Solution. We have, $\frac{d^2y}{dx^2} + y = \tan x$

$$(D^2 + 1)y = \tan x$$

A.E. is $m^2 = -1$ or $m = \pm i$

C. F. $y = A \cos x + B \sin x$

Here, $y_1 = \cos x,$ $y_2 = \sin x$

$$y_1 . y'_2 - y'_1 . y_2 = \cos x (\cos x) - (-\sin x) \sin x = \cos^2 x + \sin^2 x = 1$$

P. I. $= u. y_1 + v. y_2$ where

$$u = \int \frac{-y_2 \tan x}{y_1 \cdot y'_2 - y'_1 \cdot y_2} dx = -\int \frac{\sin x \tan x}{1} dx$$

$$= -\int \frac{\sin^2 x}{\cos x} dx = -\int \frac{1-\cos^2 x}{\cos x} dx$$

$$= \int (\cos x - \sec x)\, dx = \sin x - \log(\sec x + \tan x)$$

$$v = \int \frac{y_1 \tan x}{y_1 \cdot y'_2 - y'_1 \cdot y_2} dx = \int \frac{\cos x \cdot \tan x}{1} dx = \int \sin x\, dx = -\cos x$$

$$\text{P. I.} = u.\, y_1 + v.\, y_2$$

$$= [\sin x - \log(\sec x + \tan x)] \cos x - \cos x \sin x$$

$$= -\cos x \log(\sec x + \tan x)$$

Complete solution is

$$y = A \cos x + B \sin x - \cos x \log(\sec x + \tan x)$$ **Ans.**

Example *Use variation of parameters method to solve $y'' + y = \sec x$*

Solution. We have, $\dfrac{d^2y}{dx^2} + y = \sec x$ ... (1)

A. E. is $(m^2 + 1) = 0 \Rightarrow m^2 = -1 \Rightarrow m = \pm i$

Complementary Function of (1) is

$$\text{C. F.} = A\cos x + B\sin x$$

Here $y_1 = \cos x, \quad y_2 = \sin x$

$$\text{P. I.} = uy_1 + vy_2 \qquad \text{... (2)}$$

where $u = \displaystyle\int \frac{-y_2 \sec x}{y_1 \cdot y_2' - y_1' \cdot y_2}\, dx$ $\left[\begin{aligned} y_1 y'_2 - y_1' y_2 &= \cos x \cos x - (-\sin x)\sin x \\ &= \cos^2 x + \sin^2 x = 1 \end{aligned}\right]$

On putting the values of y_2 and $y_1 y_2' - y_1' y_2$, we get

$$u = \int \frac{-\sin x \sec x}{1}\, dx = -\int \tan x\, dx = \log \cos x$$

$$v = \int \frac{y_1 \sec x}{y_1 y_2' - y_1' y_2} dx$$

On putting the values of y_1 and $y_1 y_2' - y_1' y_2$, we get

$$v = \int \frac{\cos x \cdot \sec x}{1} dx = \int dx = x$$

Putting the values of u and v in (2), we get

$$P.I. = \cos x \cdot \log \cos x + x \sin x$$

Complete solution is $y = C.F. + P.I.$

$\Rightarrow \qquad y = A \cos x + B \sin x + \cos x \log \cos x + x \sin x$ **Ans.**

Example . *Obtain general solution of the differential equation* $x^2 y'' + xy' - y = x^3 e^x$.

Solution. The given differential equation is $x^2 y'' + xy' - y = x^3 e^x$

$$\Rightarrow \qquad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^3 e^x \qquad \dots (1)$$

Putting $x = e^z$ $\quad \Rightarrow \quad D = \frac{d}{dz}, x\frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y,$ in (1), we get

$$D(D-1)y + Dy - y = e^{3z} e^{e^z} \quad \Rightarrow \quad (D^2 - 1)y = e^{3z} e^{e^z}$$

A. E. is $m^2 - 1 = 0 \quad \Rightarrow \quad m = \pm 1$

$\therefore$ $\quad$ C. F. $= c_1 e^z + c_2 e^{-z}$

$= uy_1 + vy_2$, where $y_1 = e^{-z}$, $y = \left[y_1 = x_1 \; y_2 = \frac{1}{x} \right]$

P.I. $= uy_1 + vy_2$

Also $u = -\int \frac{y_2 z}{y_1 y_2' - y_1' y_2}\, dz = -\int \frac{e^{-z} . e^{3z} . e^{e^z}}{e^z\,(-e^{-z}) - e^z\,(e^{-z})}\, dz = -\int \frac{e^{2z} e^{e^z}}{-1-1}\, dz$

$$= \frac{1}{2}\int e^{2z} e^{e^z}\, dz = \int x^2 e^x \frac{dx}{x} = \frac{1}{2}\int x e^x dx \qquad \left[\begin{array}{l} x = e^z,\ dx = e^z dz \\ dz = \dfrac{dx}{e^z} = \dfrac{dx}{x} \end{array}\right.$$

$$= \frac{1}{2}\left[xe^x - (1)\, e^x\right] = \frac{1}{2}\,(xe^x - e^x)$$

and $v = \int \frac{y_1 z}{y_1 y_2' - y_1' y_2}\, dz = \int \frac{e^z . e^{3z} . e^{e^z}}{e^z(-e^{-z}) - e^z(e^{-z})}\, dz = \int \frac{e^{4z} e^{e^z}}{-1-1}\, dz = \int \frac{x^4 e^x}{-2} \frac{dx}{x} = -\frac{1}{2}\int x^3 e^x dx$

$$= -\frac{1}{2}[x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x]$$

P.I. $= uy_1 + vy_2 = \frac{1}{2}\,(xe^x - e^x)\, x - \frac{1}{2}\,(x^3 e^x - 3x^2 e^x + 6xe^x - 6e^x)\frac{1}{x}$

$$= \frac{1}{2}\left[x^2 - x - x^2 + 3x - 6 + \frac{6}{x}\right] e^x = \frac{1}{2}\left(2x - 6 + \frac{6}{x}\right) e^x = \left(x - 3 + \frac{3}{x}\right) e^x$$

Complete solution = C.F. + P.I.

$$y = (c_1 e^z + c_2 e^{-z}) + \left(x - 3 + \frac{3}{x}\right) e^x$$

$$= c_1 x + \frac{c_2}{x} + \left(x - 3 + \frac{3}{x}\right) e^x$$

Ans.

Example *Solve by method of variation of parameters:*

$$\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$$

Solution. $$\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$$

A. E. is $(m^2 - 1) = 0$

$$m^2 = 1, \quad m = \pm 1$$

$$C.F. = C_1 e^x + C_2 e^{-x}$$

$\therefore$ $$P.I. = uy_1 + vy_2$$

Here, $$y_1 = e^x, \quad y_2 = e^{-x}$$

and $$y_1 \cdot y'_2 - y'_1 \cdot y_2 = -e^x \cdot e^{-x} - e^x . e^{-x} = -2$$

$$u = \int \frac{-y_2 X}{y_1 . y'_2 - y'_1 \cdot y_2} dx = -\int \frac{e^{-x}}{-2} \times \frac{2}{1+e^x} dx$$

$$= \int \frac{e^{-x}}{1+e^{x}}\,dx = \int \frac{dx}{e^{x}\,(1+e^{x})} = \int \left(\frac{1}{e^{x}} - \frac{1}{1+e^{x}}\right) dx$$

$$= \int e^{-x}\,dx - \int \frac{e^{-x}}{e^{-x}+1}\,dx = -e^{-x} + \log\,(e^{-x}+1)$$

$$v = \int \frac{y_1 X}{y_1 . y'_2 - y'_1 . y_2}\,dx = \int \frac{e^{x}}{-2}\,\frac{2}{1+e^{x}}\,dx = -\int \frac{e^{x}}{1+e^{x}}\,dx = -\log\,(1+e^{x})$$

$$\text{P.I.} = u.\, y_1 + v.\, y_2 = [\,-e^{-x} + \log\,(e^{-x}+1)\,]\,e^{x} - e^{-x}\log\,(1+e^{x})$$

$$= -1 + e^{x}\log\,(e^{-x}+1) - e^{-x}\log\,(e^{x}+1)$$

Complete solution = $y = C_1 e^{x} + C_2 e^{-x} - 1 + e^{x}\log\,(e^{-x}+1) - e^{-x}\log\,(e^{x}+1)$ **Ans.**

EXERCISE

Solve the following equations by variation of parameters method.

1. $\frac{d^2y}{dx^2} - 4y = e^{2x}$ **Ans.** $y = C_1 e^{2x} + C_2 e^{-2x} + \frac{x}{4} e^{2x} - \frac{e^{2x}}{16}$

2. $\frac{d^2y}{dx^2} + y = \sin x$ **Ans.** $y = C_1 \cos x + C_2 \sin x - \frac{x}{2} \cos x + \frac{1}{4} \sin x$

3. $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = \sin x$ **Ans.** $y = C_1 e^{x} + C_2 e^{2x} + \frac{1}{10}(3 \cos x + \sin x)$

4. $\frac{d^2y}{dx^2} + y = \sec x \tan x$ **Ans.** $y = C_1 \cos x + C_2 \sin x + x \cos x + \sin x \log \sec x - \sin x$

Series Solutions of Second Order Differential Equations

INTRODUCATION

We have already studied how to find the solution of a differential equation with constant coefficients.

There are many differential equations with variable coefficients like Bessels equation, Legender's equation, Hermite's equations, Laguerrer's differential equation whose solution are not the combination of elementary functions.

The solutions are infinite series. In this chapter we will solve the differential equations by power series method and Frobenius method (extended power series method).

POWER SERIES SOLUTIONS OF DIFFERENTIAL EQUATIONS

We know that the solution of the differential equation

$$\frac{d^2y}{dx^2} - y = 0$$

are $$y = e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \quad \text{and} \quad y = e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots$$

These are power series solution of the given differential equation.

Analytic Function

JECRC UNIVERSITY
BUILD YOUR WORLD

ANALYTIC FUNCTION

A function $f(x)$ which can be expended in Taylor's series on interval containing the point x_0. The series converges to $f(x)$ for all x in the interval of convergence.

ORDINARY POINT

Consider the equation

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$$

where P, Q are polynomials in x.

$x = a$ is an ordinary point of the above equation if the denominators of P and Q do not vanish for $x = a$. *i.e.* $(P \neq \infty, Q \neq \infty)$

For example :

(*i*) $$(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0 \Rightarrow \frac{d^2y}{dx^2} + \frac{x}{1+x^2}\frac{dy}{dx} - \frac{1}{1+x^2}y = 0$$

Here, denominator of P and Q *i.e.*, $1 + x^2$ is not equal to zero at $x = 0$. Therefore $x = 0$ is an ordinary point for this differential equation.

(*ii*) $x^2\frac{d^2y}{dx^2}+(2x^2-x)\frac{dy}{dx}+y=0 \Rightarrow \frac{d^2y}{dx^2}+\frac{2x^2-x}{x^2}\frac{dy}{dx}+\frac{1}{x^2}y=0$

Here, the denominator of P and Q *i.e.*, x^2 is equal to zero for $x = 0$ ($P = \infty$, $Q = \infty$). So, $x = 0$ is not ordinary point for this equation.

WORKING RULE TO SOLVE A DIFFERENTIAL EQUATION IF $X = 0$ IS AN ORDINARY POINT OF THE EQUATION

Step 1. Assume the solution $y = \Sigma a_k x^k$ *i.e.*,

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...$$

Step 2. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ of step (1).

Step 3. Substitute the values of $y, \frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in the given differential equation.

Step 4. Equate to zero the coefficients of various power of x, to find out a_2, a_3 ..... in terms of a_0 and a_1.

Step 5. Equate to zero the coefficient of x^k to get a recurrence relation of a's.

Step 6. Substitute $k = 0, 1, 2, 3$, ... in the recurrence relations to get the values of a_2, a_3, ... a_n, ...

Step 7. Substitute the values of a_2, a_3, ... etc (obtained in step 6) in the solution (1)

When $x = 0$ is the ordinary point.

Solved Example

Example 1. *Solve in series the equation* $\dfrac{d^2y}{dx^2} + x^2 y = 0.$

Solution. We have, $\dfrac{d^2y}{dx^2} + x^2 y = 0$ ... (1)

The denominator of P and Q is not zero so $x = 0$ is the ordinary point.

Let $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + a_7 x^7 + a_8 x^8 + \ldots + a_n x^n + \ldots$ ... (2)

Here $P \neq \infty$, and $Q \neq \infty$ for $x = 0$. So, $x = 0$ is the ordinary point of the equation (1).

Then $\dfrac{dy}{dx} = a_1 + 2a_2 x + 3\, a_3\, x^2 + 4\, a_4\, x^3 + 5\, a_5\, x^4 + 6\, a_6\, x^5 + 7\, a_7\, x^6 + 8\, a_8\, x^7 + \ldots$

$$\frac{d^2y}{dx^2} = 2a_2 + 2\,.\,3\, a_3\, x + 3\,.\,4\, a_4\, x^2 + 4\,.\,5\, a_5\, x^3 + 5\,.\,6\, a_6\, x^4 + 6\,.\,7\, a_7\, x^5 + 7\,.\,8\, a_8\, x^6 + \ldots$$

Substituting the values of $\dfrac{d^2y}{dx^2}$ and y in (1), we get

$$2\, a_2 + 2.3\, a_3\, x + 3.4\, a_4\, x^2 + 4.5\, a_5\, x^3 + 5.6\, a_6\, x^4 + 6.7\, a_7\, x^5 + 7.8\, a_8\, x^6 + \ldots$$
$$+ x^2 (a_0 + a_1\, x + a_2\, x^2 + a_3\, x^3 + a_4\, x^4 + a_5\, x^5 + a_6\, x^6 + a_7\, x^7 + \ldots) = 0$$

$$2\, a_2 + 6\, a_3\, x + (a_0 + 12\, a_4)\, x^2 + (a_1 + 20 a_5)\, x^3 + (a_2 + 30\, a_6)\, x^4 + \ldots$$
$$+ [a_{n-2} + (n+2)(n+1)\, a_{n+2}]\, x^n + \ldots = 0$$

Equating to zero the coefficients of the various powers of x, we obtain

$$a_2 = 0, \quad a_3 = 0$$

$$a_0 + 12a_4 = 0 \quad i.e. \quad a_4 = -\frac{1}{12}a_0$$

$$a_1 + 20\, a_5 = 0 \quad i.e. \quad a_5 = -\frac{1}{20}a_1$$

$$a_2 + 30\, a_6 = 0 \quad i.e. \quad a_6 = -\frac{1}{30}\, a_2 = 0 \qquad (a_2 = 0)$$

and so on. In general

$$a_{n-2} + (n+2)(n+1)\, a_{n+2} = 0 \qquad \Rightarrow \qquad \boxed{a_{n+2} = -\frac{a_{n-2}}{(n+1)(n+2)}}$$

Putting $n = 5$, $$a_7 = -\frac{a_3}{6 \times 7} = 0 \qquad (a_3 = 0)$$

Putting $n = 6$, $$a_8 = -\frac{a_4}{7 \times 8} = \frac{a_0}{12 \times 7 \times 8}$$

Putting $n = 7$, $$a_9 = -\frac{a_5}{8 \times 9} = \frac{a_1}{20 \times 8 \times 9}$$

Putting $n = 8$, $$a_{10} = -\frac{a_6}{9 \times 10} = 0, \qquad (a_6 = 0)$$

Putting $n = 9$, $$a_{11} = -\frac{a_7}{11 \times 10} = 0, \qquad (a_7 = 0)$$

Example 2

Putting $n = 10$, $$a_{12} = -\frac{a_8}{12\times 11} = -\frac{a_0}{12\times 8\times 7\times 11\times 12}$$

Substituting these values in (2), we get

$$y = a_0 + a_1 x - \frac{1}{12}a_0 x^4 - \frac{a_1}{20}x^5 + \frac{a_0}{12\times 7\times 8}x^8 + \frac{a_1}{20\times 8\times 9}x^9 - \frac{a_0}{12\times 8\times 7\times 11\times 12}x^{12} + \ldots$$

$$y = a_0\left(1 - \frac{1}{12}x^4 + \frac{x^8}{12\times 7\times 8} - \frac{x^{12}}{12\times 8\times 7\times 11\times 12} + \ldots\right) + a_1\left(x - \frac{x^5}{20} + \frac{x^9}{20\times 8\times 9} - \ldots\right)$$ **Ans.**

Example 2. *Solve the following differential equation in series*

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 4y = 0.$$ *(U.P. II Semester summer 2006)*

Solution. We have,

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 4y = 0 \Rightarrow \frac{d^2y}{dx^2} - \frac{x}{1-x^2}\frac{dy}{dx} + \frac{4}{1-x^2}y = 0 \quad \ldots (1)$$

Here, $P \neq \infty$ and $Q \neq \infty$ for $x = 0$. So, $x = 0$ is an ordinary point of the given equation.

Assume the solution

$$y = \Sigma a_k x^k$$

$$\Rightarrow \quad y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_n x^n + \ldots\ldots \quad \ldots (2)$$

$$y' = \Sigma a_k (k) x^{k-1}$$

$$y'' = \Sigma a_k (k)(k-1) x^{k-2}$$

Contd…

Putting these values in the given equation, we get

$$(1-x^2)\Sigma a_k(k)(k-1)x^{k-2} - x\Sigma a_k(k)x^{k-1} + 4\Sigma a_k x^k = 0$$

$$\Rightarrow \quad \Sigma a_k(k)(k-1)x^{k-2} - \Sigma a_k(k)(k-1)x^k - \Sigma a_k(k)x^k + 4\Sigma a_k x^k = 0$$

$$\Rightarrow \quad \Sigma a_k(k)(k-1)x^{k-2} - \Sigma a_k x^k\{k(k-1)+k-4\} = 0$$

$$\Rightarrow \quad \Sigma a_k k(k-1)x^{k-2} - \Sigma a_k x^k(k^2-4) = 0$$

Now, equating the coefficient of x^k equal to zero. By putting $k+2$ for k in first summation and $k = k$ in second summation, we have

$$a_{k+2}(k+2)(k+1) - a_k(k^2-4) = 0$$

$$a_{k+2}(k+2)(k+1) = a_k(k^2-4)$$

$$a_{k+2} = \frac{k^2-4}{(k+2)(k+1)}a_k = \frac{k-2}{k+1}a_k \qquad \Rightarrow \qquad \boxed{a_{k+2} = \frac{k-2}{k+1}a_k}$$

If $k = 0$, $\quad a_2 = -2a_0$

If $k = 1$, $\quad a_3 = -\frac{1}{2}a_1$

If $k = 2$, $\quad a_4 = \frac{0}{3}a_2 = 0$

If $k = 3$, $\quad a_5 = \frac{1}{4}a_3 = \frac{1}{4}\left(-\frac{1}{2}\right)a_1 = -\frac{a_1}{8}$

If $k = 4$, $\quad a_6 = \frac{2}{5}a_4 = \frac{2}{5}\times 0 = 0$

and so on.

Example 3

Substituting these values is (2), we get

$$y = a_0 + a_1 x + (-2a_0)x^2 + \left(-\frac{1}{2}a_1\right)x^3 + 0x^4 + \left(\frac{-a_1}{8}\right)x^5 + 0.x^6 + \ldots\ldots$$

$$= a_0 + a_1 x - 2a_0 x^2 - \frac{1}{2}a_1 x^3 - \frac{a_1}{8}x^5 + \ldots\ldots$$

$$= a_0(1-2x^2) + a_1 x\left(1 - \frac{x^2}{2} - \frac{x^4}{8} + \ldots\ldots\right)$$

Ans.

Example 3. *Find the power series solution of* $(1 - x^2)\ y'' - 2xy' + 2y = 0$ *about* $x = 0$.

(A.M.I.E.T.E., Winter 2000)

Solution. Here, we have

$$(1 - x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0 \Rightarrow \frac{d^2y}{dx^2} - \frac{2x}{1-x^2}\frac{dy}{dx} + \frac{2}{1-x^2}y = 0$$

Here $P \neq \infty$ and $Q \neq \infty$ for $x = 0$. So $x = 0$ is an ordinary point.

Let $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_n x^n + \ldots$ be the required solution.

$$y = \sum_{k=0}^{\infty} a_k x^k$$

Then $$\frac{dy}{dx} = \sum_{k=1}^{\infty} a_k \cdot k\, x^{k-1}, \qquad \frac{d^2y}{dx^2} = \sum_{k=2}^{\infty} a_k \cdot k(k-1)\, x^{k-2}$$

Contd…

Substituting the values of $y, \dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ in the given equation, we get

$$(1-x^2)\,\Sigma a_k\, k\cdot(k-1)\,x^{k-2} - 2x\,\Sigma a_k\cdot k\,x^{k-1} + 2\,\Sigma a_k\, x^k = 0$$

$\Rightarrow$ $$\Sigma a_k\cdot k\cdot(k-1)\,x^{k-2} - \Sigma a_k k(k-1)\,x^k - 2\,\Sigma a_k . k\,x^k + 2\,\Sigma a_k x^k = 0$$

$\Rightarrow$ $$\Sigma a_k\cdot k\cdot(k-1)\,x^{k-2} - \Sigma[k(k-1)+2k-2]\,a_k x^k = 0$$

$\Rightarrow$ $$\Sigma a_k\cdot k\cdot(k-1)\,x^{k-2} - \Sigma(k^2+k-2)\,a_k x^k = 0$$

where the first summation extends over all values of k from 2 to ∞ and the second from $k = 0$ to ∞.

Now equating the coefficient of x^k equal to zero, we have

$\Rightarrow$ $$(k+2)\,(k+1)\,a_{k+2} - (k^2+k-2)\,a_k = 0$$

$\Rightarrow$ $$a_{k+2} = \frac{k^2+k-2}{(k+2)(k+1)}\,a_k = \frac{(k+2)(k-1)}{(k+2)(k+1)}\,a_k$$

$\Rightarrow$ $$\boxed{a_{k+2} = \frac{k-1}{k+1}\,a_k}$$

For $k = 0$ $$a_2 = -a_0,\ a_3 = 0,\ a_4 = \frac{a_2}{3} = -\frac{a_0}{3},\ a_5 = \frac{2}{4}a_3 = 0$$

For $k = 4$ $$a_6 = \frac{3}{5}a_4 = \frac{3}{5}\left(-\frac{a_0}{3}\right) = -\frac{a_0}{5},\ a_7 = \frac{4}{6}a_5 = 0, \text{ etc.}$$

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \ldots$$

$$= a_0 + a_1x - a_0x^2 + 0 - \frac{a_0}{3}x^4 + 0 - \frac{a_0}{5}x^6 + 0 + \ldots$$ Ans.

EXERCISE

Solve the following differential equation by power series method :

1. $\dfrac{d^2y}{dx^2}+xy=0$ **Ans.** $y=a_0\left(1-\dfrac{x^3}{3!}+\dfrac{4x^6}{6!}-\dfrac{28x^9}{9!}+\ldots\right)+a_1\left(x-\dfrac{2x^4}{4!}+\dfrac{10x^7}{7!}+\ldots\right)$

(AMIETE, June 2010)

2. $y''-xy'+x^2y=0$ **Ans.** $y=a_0\left(1-\dfrac{1}{12}x^4-\ldots\right)+a_1\left(x+\dfrac{1}{6}x^3-\dfrac{1}{40}x^5\ldots\right)$

3. $(x^2+1)y''+xy'-xy=0$ (U.P. (C.O.) 2008)

Ans. $y=a_0\left(1+\dfrac{x^3}{6}-\dfrac{3}{40}x^5+\ldots\right)+a_1\left(x-\dfrac{x^3}{6}+\dfrac{x^4}{12}+\dfrac{3}{40}x^5\ldots\right)$

Singular Points

SINGULAR POINTS ABOUT $x = a$

Definition. Consider the equation

$$y'' + P(x)\,y' + Q(x)\,y = 0 \qquad \dots (1)$$

and assume that functions P and Q are not analytic ($P = \infty$ or $Q = \infty$) at $x = a$, so that $x = a$ is not *ordinary point* but a *singular point of* (1).

There are two types of singular points. (1) Regular singular point, (2) Irregular singular points.

1. **Regular Singular Point:**
   If $(x - a)\,P$ and $(x - a)^2\,Q$ are not infinite at $x = a$, then $x = a$ is a regular singular point.
2. **Irregular Singular Point:**
   If $(x - a)\,P$ and $(x - a)^2\,Q$ are infinite at $x = a$, then $x = a$ is an irregular singular point.

Contd…

SINGULAR POINT ABOUT x = 0.

Consider the differential equation

$$\frac{d^2y}{dx^2}+P\frac{dy}{dx}+Qy=0$$

If $(x-0)\,P$ and $(x-0)^2\,Q$ are not infinite at $x = 0$, then $x = 0$ is a regular singular point. Otherwise it is an irregular singular point.

Note: In this section we will solve those differential equation where x_0 is a regular singular point.

Example 5. *Find regular singular points of the differential equation.*

$$2x^2\frac{d^2y}{dx^2}+3x\frac{dy}{dx}+(x^2-4)\,y=0 \qquad \ldots (1)$$

Solution. We have,
$$\frac{d^2y}{dx^2}+\frac{3}{2x}\frac{dy}{dx}+\frac{x^2-4}{2x^2}\,y=0$$

$$P=\frac{3}{2x} \quad \text{and } Q=\frac{x^2-4}{2x^2}$$

P and Q are not analytic (infinity) at $x = 0$. So, $x = 0$ is not ordinary point but as $(x-0)\,P$ and $(x-0)^2\,Q$ are analytic (not infinite) so $x = 0$ is a regular singular point.

$$x\cdot P=x\left(\frac{3}{2x}\right)=\frac{3}{2} \neq \infty \text{ at } x=0.$$

$$x^2Q=x^2\cdot\frac{x^2-4}{2x^2}=\frac{1}{2}(x^2-4) \neq \infty \text{ at } x=0$$ **Ans.**

Solved Example

Example 6. *Find regular singular points of the differential equation.*

$$x(x-2)^2 y'' + 2(x-2)y' + (x+3)y = 0 \qquad \dots (1)$$

Solution. Here, we have $P = \dfrac{2(x-2)}{x\ (x-2)^2} = \dfrac{2}{x\ (x-2)}$ and $Q = \dfrac{x+3}{x\,(x-2)^2}$

P and Q are not analytic ($P = \infty$, $Q = \infty$) at $x = 0$ and $x = 2$.
Hence both these points are singular points of (1).
(*i*) At $x = 0$

$$xP = x \cdot \frac{2}{x\,(x-2)} = \frac{2}{x-2} \neq \infty \quad \text{at } x = 0$$

$$x^2 Q = x^2 \cdot \frac{x+3}{x\,(x-2)^2} = \frac{x\,(x+3)}{(x-2)^2} \neq \infty \quad \text{at } x = 0$$

Hence, $x\,P$ and $x^2\,Q$ are analytic ($xP \neq \infty$, $x^2P \neq \infty$) at $x = 0$. So $x = 0$ is a regular singular point.

(*ii*) At $x = 2$

$$(x-2)P = (x-2)\cdot\frac{2}{x(x-2)} = \frac{2}{x} \neq \infty \text{ at } x = 2$$

$$(x-2)^2 Q = (x-2)^2 \frac{(x+3)}{x(x-2)^2} = \frac{x+3}{x} \neq \infty \text{ at } x = 2.$$

Since both $(x-2)\,P$ and $(x-2)^2Q$ are analytic ($(x-2)\,P \neq \infty$, $(x-2)^2Q \neq 0$) at $x = 2$, so $x = 2$ is a regular singular point.

FROBENIUS METHOD

If $x = 0$ is a regular singularity of the equation.

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)\,y = 0 \qquad [P_0(0) = 0] \quad \ldots (1)$$

Then the series solution is

$$\boxed{y = x^m (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots) = \sum_{k=0}^{\infty} a_k x^{m+k}}$$

The value of m will be determined by substituting the expressions for $y, \frac{dy}{dx}, \frac{d^2y}{dx^2}$ in (1), we get the identity.

On equating the coefficient of lowest power of x in the identity to zero, a quadratic equation in m (**indicial equation**) is obtained.

Thus, we will get two values of m. The series solution of (1) will depend on the nature of the roots of the indicial equation.

(*i*) **Case 1 : When roots m_1, m_2 are distinct and not differing by an integer i.e.** $m_1 - m_2 \neq 0$ or a positive integer. $e.g.,\ m_1 = \frac{1}{2}, m_2 = 2$.

The complete solution is $\boxed{y = c_1 (y)_{m_1} + c_2 (y)_{m_2}}$

Contd…

(*ii*) **Case 2 : When roots m_1, m_2 are equal i.e. $m_1 = m_2$**

$$\boxed{y = c_1(y)_{m_1} + c_2\left(\frac{\partial y}{\partial m}\right)_{m_1}}$$

(*iii*) **Case 3 : When roots m_1, m_2 are distinct and differ by an integer** $(m_1 < m_2)$

e.g., $m_1 = \frac{3}{2}$, $m_2 = \frac{5}{2}$ or $m_1 = 2$, $m_2 = 4$.

If some of the coefficients of y series become infinite when $m = m_1$, to overcome this difficulty, replace a_0 by $b_0\,(m - m_1)$. We get a solution which is only a constant multiple of the first solution.

$$\boxed{a_0 = b_0(m - m_1)}$$

Complete solution is $\boxed{y = c_1(y)_{m_1} + c_2\left(\frac{\partial y}{\partial m}\right)_{m_1}}$

(*iv*) **Case 4 : Roots are distinct and differing by an integer, making some coefficient indeterminate**

Complete solution is $y = c_1(y)_{m_1} + c_2(y)_{m_2}$

if the coefficients do not become infinite when $m_1 = m_2$.

Example of Case - I

Case I : When the roots are distinct and not differing by an integer.

Example 7. *Find solution in generalized series form about $x = 0$ of the differential equation*

$$3x\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=0$$

Solution. We have, $$3x\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=0 \qquad \text{... (1)}$$

Here, $xP(x)$ and $x^2Q(x)$ are analytic (not infinite). So, $x = 0$ is a regular singular point, we assume the solution in the form

$$y=\sum_{k=0}^{\infty}a_k x^{m+k}$$

Such that

$$\frac{dy}{dx}=\sum_{k=0}^{\infty}a_k(m+k)\,x^{m+k-1}$$

$$\frac{d^2y}{dx^2}=\sum_{k=0}^{\infty}a_k(m+k)(m+k-1)\,x^{m+k-2}$$

Contd…

Substituting for y, $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in the given equation (1), we get

$$3\Sigma a_k (m+k)(m+k-1)\, x^{m+k-1} + 2\,\Sigma a_k\,(m+k) x^{m+k-1} + \Sigma a_k x^{m+k} = 0$$

$$\Rightarrow \quad \Sigma a_k [3(m+k)(m+k-1) + 2(m+k)] x^{m+k-1} + \Sigma a_k x^{m+k} = 0 \qquad \ldots (2)$$

The coefficient of the lowest degree term x^{m-1} in the identity (2) is obtained by putting $k = 0$ in first summation only and equating it to zero. Then the **indicial equation** is

$$a_0\,[3\,m\,(m-1) + 2\,m] = 0 \quad \Rightarrow \quad a_0\,[3\,m^2 - m] = 0 \quad \Rightarrow \quad \boxed{a_0 m(3m-1) = 0}$$

Since $\quad a_0 \neq 0, \;\; m = 0, \;$ or $\; 1/3$

The coefficient of next lowest degree term x^m in the identity (2) is obtained by putting $k = 1$ in first summation and $k = 0$ in the second summation and equating it to zero.

$$a_1\,[3\,(m+1)\,m + 2\,(m+1)] + a_0 = 0$$

$$\Rightarrow \quad a_1\,[3\,m^2 + 5m + 2] + a_0 = 0 \;\Rightarrow\; a_1\,(3m+2)\,(m+1) + a_0 = 0$$

$$\Rightarrow \quad a_1 = -\frac{1}{(3m+2)\,(m+1)}\,a_0$$

Equating to zero the coefficient of x^{m+k}, the recurrence relation is given by

$$a_{k+1}\,[3\,(m+k+1)\,(m+k) + 2\,(m+k+1)] + a_k = 0.$$

$$\Rightarrow \quad a_{k+1}\,(m+k+1)\,(3\,m + 3k + 2) + a_k = 0 \Rightarrow \boxed{a_{k+1} = \frac{-1}{(m+k+1)\,(3m+3k+2)}\,a_k}$$

Contd…

This gives

For $k = 0$, $$a_1 = \frac{-1}{(m+1)(3m+2)} a_0$$

For $k = 1$, $$a_2 = \frac{-1}{(m+2)(3m+5)} a_1 = \frac{1}{(m+1)(m+2)(3m+2)(3m+5)} a_0$$

For $k = 2$, $$a_3 = \frac{-1}{(m+3)(3m+8)} a_2$$

$$= \frac{-1}{(m+1)(m+2)(m+3)(3m+2)(3m+5)(3m+8)} a_0$$

For $m = 0$

$$a_1 = -\frac{1}{2} a_0, \qquad a_2 = \frac{1}{20} a_0, \qquad a_3 = -\frac{1}{480} a_0$$

Hence, for $m = 0$, $y_1 = a_0\left(1 - \frac{1}{2}x + \frac{1}{20}x^2 - \frac{1}{480}x^3 + \dots\right)$

For $m = \frac{1}{3}$

$$a_1 = -\frac{1}{4} a_0, \qquad a_2 = \frac{1}{56} a_0, \qquad a_3 = \frac{-1}{1680} a_0$$

Hence for $m = \frac{1}{3}$, the second solution is

$$y_2 = a_0\left(x^{\frac{1}{3}} - \frac{1}{4}x^{\frac{4}{3}} + \frac{1}{56}x^{\frac{7}{3}} - \frac{1}{1680}x^{\frac{10}{3}} + \dots\right)$$

Example of Case - II

Thus the complete solution is

$$y = Ay_1 + By_2$$

$$y = a_0\left(1 - \frac{x}{2} + \frac{x^2}{20} - \frac{x^3}{480} + \ldots\right) + b_0 x^{1/3}\left(1 - \frac{x}{4} + \frac{x^2}{56} - \frac{x^3}{1680} + \ldots\right) \quad \textbf{Ans.}$$

Case II. When the roots of indicial equation are equal.

Example 11. *Solve* $x(x-1)y'' + (3x-1)y' + y = 0$

Solution. $x(x-1)y'' + (3x-1)y' + y = 0 \quad \ldots (1)$

Here $xP(x)$ and $x^2Q(x)$ are analytic (Not infinite) at $x = 0$. So, $x = 0$ is a regular singular point, we assume the solution in the form

$$y = \sum_{k=0}^{\infty} a_k x^{m+k}$$

such that

$$\frac{dy}{dx} = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1}$$

$$\frac{d^2y}{dx^2} = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2}$$

Substituting the expressions for $y, \frac{dy}{dx}, \frac{d^2y}{dx^2}$ in (1), we have

$$x(x-1) \Sigma a_k (m+k)(m+k-1) x^{m+k-2} + (3x-1) \Sigma a_k (m+k) x^{m+k-1} + \Sigma a_k x^{m+k} = 0$$

Contd…

$$\Rightarrow \quad \Sigma a_k (m+k)(m+k-1) x^{m+k} - \Sigma a_k (m+k)(m+k-1) x^{m+k-1}$$
$$+ 3\, \Sigma a_k (m+k) x^{m+k} - \Sigma a_k (m+k) x^{m+k-1} + \Sigma a_k x^{m+k} = 0$$

$$\Rightarrow \quad \Sigma a_k [(m+k)(m+k-1) + 3(m+k) + 1] x^{m+k}$$
$$- \Sigma a_k [(m+k)(m+k-1) + (m+k)] x^{m+k-1} = 0$$

$$\Rightarrow \quad \Sigma a_k [(m+k)(m+k+2) + 1] x^{m+k} - \Sigma a_k (m+k)^2 x^{m+k-1} = 0 \qquad \dots (2)$$

The coefficient of lowest degree term x^{m-1} in (2) is obtained by putting $k = 0$ in the second summation only of (2) and equating it to zero. Then the **indicial equation** is

$$\boxed{a_0 (m+0)^2 = 0} \quad \Rightarrow \quad m = 0,\ 0 \ \text{as}\ a_0 \neq 0$$

The coefficient of the next lowest degree term x^m in (2) is obtained by putting $k = 0$ in the first summation and $k = 1$ in the second summation only of (2) and equating it to zero, we get

$$a_0 [(m+0)(m+2) + 1] - a_1 (m+1)^2 = 0$$

$$\Rightarrow \quad a_0 (m^2 + 2m + 1) - a_1 (m^2 + 2m + 1) = 0$$

$$a_1 - a_0 = 0 \quad \Rightarrow \quad a_1 = a_0 \quad (\text{as } m = 0)$$

Equating the coefficient of x^{m+k} to zero, the recurrence relation is given by

$$a_k [(m+k)(m+k+2) + 1] - a_{k+1} (m+k+1)^2 = 0$$
$$a_k (m+k+1)^2 - a_{k+1} (m+k+1)^2 = 0$$

Contd…

Hence, $a_{k+1} = a_k$

$$y = x^m [a_0 + a_1 x + a_2 x^2 + ...]$$

$$y = a_0 x^m [1 + x + x^2 + x^3 + ...]$$

For $m = 0$

when $m = 0, 0$, this gives only one solution instead of two.

Second solution is given by

$$\left(\frac{\partial y}{\partial m}\right)_{m=0} \qquad \text{and} \qquad y_1 = a_0 (1 + x + x^2 + x^3)$$

$$\frac{\partial y}{\partial m} = a_0 x^m \log x [1 + x + x^2 + x^3 + ...]$$

$$y_2 = a_0 \log x [1 + x + x^2 + x^3 + ...] \qquad m = 0$$

$$y_1 = a_0 [1 + x + x^2 + x^3 + ...] \qquad m = 0$$

$$y = A y_1 + B y_2$$

$$y = A [1 + x + x^2 + x^3 + ...] + B \log x (1 + x + x^2 + x^3 + ...) \qquad \textbf{Ans.}$$

Example of Case - III

Case III : When m_1 and m_2 are distinct and differing by an integer, then

$$y = c_1 (y)_{m_1} + c_2 \left(\frac{\partial y}{\partial m}\right)_{m_2} \qquad \left[\begin{array}{r}\text{If coefficient} = \infty \\ \text{when } m = m_2\end{array}\right]$$

Example 14. *Solve*

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 4) y = 0 \qquad \text{... (1)}$$

Solution. Here $xP(x)$ and $x^2 Q(x)$ are analytic (not infinite) at $x = 0$. So, $x = 0$ is regular singular point of this equation.

Let $\quad y = \Sigma a_k x^{m+k}$

$$\frac{dy}{dx} = \Sigma a_k (m+k) x^{m+k-1}$$

$$\frac{d^2 y}{dx^2} = \Sigma a_k (m+k)(m+k-1) x^{m+k-2}$$

Substituting the values of $\frac{d^2 y}{dx^2}, \frac{dy}{dx}$ and y in (1), we get

$$x^2 \; \Sigma a_k (m+k)(m+k-1) x^{m+k-2} + x \Sigma a_k (m+k) x^{m+k-1} + (x^2 - 4) \; \Sigma a_k x^{m+k} = 0$$

$$\Rightarrow \qquad \Sigma a_k [(m+k)(m+k-1) + (m+k) - 4] x^{m+k} + \Sigma a_k x^{m+k+2} = 0$$

$$\Rightarrow \qquad \Sigma a_k (m+k+2)(m+k-2) x^{m+k} + \Sigma a_k x^{m+k+2} = 0 \qquad \text{... (2)}$$

Contd…

The coefficient of lowest degree term x^m in (2) is obtained by putting $k = 0$ in first summation only and equating it to zero. Then the indicial equation is

$$\boxed{a_0(m+2)\ (m-2) = 0} \Rightarrow m = 2, -2$$

The coefficient of next lowest term x^{m+1} in (2) is obtained by putting $k = 1$ in first summation only and equating it to zero.

$$a_1(m+3)(m-1) = 0 \Rightarrow a_1 = 0$$

Equating to zero the coefficient of x^{m+k+2}, we get

$$a_{k+2}(m+k+4)(m+k) + a_k = 0 \quad \Rightarrow \quad \boxed{a_{k+2} = -\frac{a_k}{(m+k+4)(m+k)}}$$

$$a_1 = a_3 = a_5 = \ldots = 0$$

$$a_2 = -\frac{a_0}{m(m+4)}$$

$$a_4 = -\frac{a_2}{(m+2)(m+6)} = \frac{a_0}{m(m+2)(m+4)(m+6)}$$

$$a_6 = -\frac{a_4}{(m+4)(m+8)} = -\frac{a_0}{m(m+2)(m+4)^2(m+6)(m+8)}$$

Hence

$$y = a_0 x^m \left[1 - \frac{x^2}{m(m+4)} + \frac{x^4}{m(m+2)(m+4)(m+6)} - \frac{x^6}{m(m+2)(m+4)^2(m+6)(m+8)} + \ldots\right] \quad \ldots (3)$$

Contd…

Putting m = 2 in (3), we get

$$y_1 = a_0 x^2 \left[1 - \frac{x^2}{2\times 6} + \frac{x^4}{2\times 4\times 6\times 8} - \frac{x^6}{2\times 4\times 6^2\times 8\times 10} + \dots\right] \quad \dots (4)$$

For m = – 2

Coefficient of x^4, x^6 etc. in (3) becomes infinite on putting m = –2. To overcome this difficulty, we put

$a_0 = b_0(m+2)$ in (1) and we get

$$y = b_0 x^m \left[(m+2) - \frac{(m+2)x^2}{m(m+4)} + \frac{x^4}{m(m+4)(m+6)} - \frac{x^6}{m(m+4)^2(m+6)(m+8)} + \dots\right] \quad \dots (5)$$

On differentiating (5) w.r.t. 'm', we get

$$\frac{\partial y}{\partial m} = b_0(x^m \cdot \log x)\left[(m+2) - \frac{(m+2)x^2}{m(m+4)} + \frac{x^4}{m(m+4)(m+6)} - \frac{x^6}{m(m+4)^2(m+6)(m+8)} + \dots\right]$$

$$+ b_0 x^m \left[1 - \frac{(m+2)x^2}{m(m+4)}\left(\frac{1}{m+2} - \frac{1}{m} - \frac{1}{m+4}\right) + \frac{x^4}{m(m+4)(m+6)}\left(-\frac{1}{m} - \frac{1}{m+4} - \frac{1}{m+6}\right) + \dots\right]$$

On replacing m by – 2, we get

$$\left(\frac{\partial y}{\partial m}\right)_{m=-2} = (b_0 x^{-2}\log x)\left[0 - 0 + \frac{x^4}{(-2)(2)(4)} - \frac{x^6}{(-2)(2)^2(4)(6)} + \dots\right]$$

$$+ b_0 x^{-2}\left[1 + \frac{x^2}{2^2} + \frac{x^4}{2^2\times 4}\left(\frac{1}{4}\right) + \dots\right]$$

Contd…

$$\Rightarrow \quad y_2 = b_0 x^2 \log x\left(-\frac{1}{2^2\times 4}+\frac{x^2}{2^3\times 4\times 6}-\frac{x^4}{2^3\times 4^2\times 6\times 8}+\dots\right)+b_0 x^{-2}\left(1+\frac{x^2}{2^2}+\frac{x^4}{2^2\times 4^2}+\dots\right)$$

General solution is $y = c_1 y_1 + c_2 y_2$

$$y = c_1 x^2\left(1-\frac{x^2}{2\times 6}+\frac{x^4}{2\times 4\times 6\times 8}-\frac{x^6}{2\times 4\times 6^2\times 8\times 10}+\dots\right)$$

$$+c_2\left[x^2 \log x\left(-\frac{1}{2^2\times 4}+\frac{x^2}{2^3\times 4\times 6}-\frac{x^4}{2^3\times 4^2\times 6\times 8}+\dots\right)+x^{-2}\left(1+\frac{x^2}{2^2}+\frac{x^4}{2^2\times 4^2}+\dots\right)\right]$$

Ans.

Example of Case - IV

Case IV. If the roots differ by an integer such that one or more coefficients are indeterminate.

Example 17. *Find the extended power series solution of the differential equation*

$$x^2 y'' + 4xy' + (x^2 + 2)\, y = 0 \qquad \text{... (1)}$$

Solution. Here $xP(x)$ and $x^2Q(x)$ are analytic (not infinity) at $x = 0$. So, $x = 0$ is a regular singular point of this equation

Let $y = \Sigma\, a_k x^{m+k}$ be the required solution of the given equation.

Then $\quad \dfrac{dy}{dx} = \Sigma a_k (m+k)\, x^{m+k-1}, \qquad \dfrac{d^2y}{dx^2} = \Sigma a_k (m+k)(m+k-1)\, x^{m+k-2}$

Substituting the values of y, $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ in the given equation, we get

$$x^2\, \Sigma a_k (m+k)(m+k-1)\, x^{m+k-2} + 4\,x\, \Sigma a_k (m+k)\, x^{m+k-1} + (x^2+2)\, \Sigma a_k x^{m+k} = 0$$

$$\Sigma\, a_k (m+k)(m+k-1)\, x^{m+k} + 4\, \Sigma\, a_k (m+k)\, x^{m+k} + \Sigma\, a_k x^{m+k+2} + \Sigma\, 2a_k x^{m+k} = 0$$

$$\Sigma\, a_k\, [(m+k)(m+k-1) + 4(m+k) + 2]\, x^{m+k} + \Sigma a_k x^{m+k+2} = 0$$

$$\Sigma\, a_k [(m+k)^2 + 3(m+k) + 2]\, x^{m+k} + \Sigma a_k x^{m+k+2} = 0 \qquad \text{... (2)}$$

The coefficient of lowest degree term x^m in (2) is obtained by putting $k = 0$ in first summation only and equating it to zero. Then the indicial equation is

$$\boxed{\boldsymbol{a_0(m^2 + 3m + 2) = 0}}$$

$$a_0 \neq 0,\ m^2 + 3m + 2 = 0 \quad \Rightarrow \quad (m+1)(m+2) = 0 \quad \Rightarrow \quad m = -1, -2$$

Contd…

The coefficient of next lowest degree term x^{m+1} in (2) is obtained by putting $k = 1$ in first summation only and equating it to zero.

$$a_1[m^2 + 5m + 6] = 0 \text{ or } a_1(m+2)(m+3) = 0 \Rightarrow a_1 = \frac{0}{(m+2)(m+3)}$$

when $m = -2$, a_1 becomes indeterminate $\left(\frac{0}{0}\right)$. When $m = -2$ we get the identity $a_1\,(0) = 0$ which is satisfied by every value of a_1. Therefore in this case we can take a_1 as arbitrary constant.

Equating to zero the coefficient of x^{m+k+2}

$$a_{k+2}\,[(m+2+k)^2 + 3\,(m+2+k) + 2] + a_k = 0$$

$$\Rightarrow \quad a_{k+2}[m^2 + (2k+4+3)m + (k+2)^2 + 3(k+2) + 2] + a_k = 0$$

$$\Rightarrow \quad a_{k+2}[m^2 + (2k+7)m + k^2 + 7k + 12] + a_k = 0$$

$$\Rightarrow \quad \boxed{a_{k+2} = -\frac{1}{m^2 + (2k+7)m + k^2 + 7k + 12}\,a_k}$$

For $k = 0$,
$$a_2 = -\frac{1}{m^2 + 7m + 12}\,a_0 = -\frac{1}{(m+3)(m+4)}\,a_0$$

JECRC Foundation

Contd…

$k = 1$ $$a_3 = -\frac{1}{m^2+9m+20}a_1 = -\frac{1}{(m+4)(m+5)}a_1$$

$k = 2$ $$a_4 = -\frac{1}{m^2+11m+30}a_2 = \frac{1}{(m+3)(m+4)(m+5)(m+6)}a_0$$

$k = 3$ $$a_5 = -\frac{1}{m^2+13m+42}a_3 = \left\{\frac{1}{(m+4)(m+5)(m+6)(m+7)}a_1\right\}$$

For $m = -1$

$$a_2 = -\frac{1}{6}a_0, \quad a_3 = \frac{1}{12}a_1, \quad a_4 = \frac{1}{120}a_0, \quad a_5 = \frac{1}{360}a_1$$

First solution is $$y_1 = x^{-1}\left[1-\frac{1}{6}x^2+\frac{1}{120}x^4+\ldots\right]a_0+\left[1-\frac{1}{12}x^2+\frac{x^4}{360}+\ldots\right]a_1$$

For $m = -2$

$$a_2 = -\frac{1}{2}a_0, \quad a_3 = -\frac{1}{6}a_1, \quad a_4 = \frac{1}{24}a_0, \quad a_5 = \frac{1}{120}a_1$$

Second solution is $$y_2 = x^{-2}\left[1-\frac{x^2}{2}+\frac{x^4}{24}+\ldots\right]a_0+\left[\frac{1}{x}-\frac{x}{6}+\frac{x^3}{120}+\ldots\right]a_1$$

$$y_2 = x^{-2}\left[\left\{1-\frac{x^2}{2}+\frac{x^4}{24}+\ldots\right\}a_0+\left\{x-\frac{x^3}{6}+\frac{x^5}{120}+\ldots\right\}a_1\right]$$

$$y_2 = x^{-2}\left[a_0\cos x + a_1 \sin x\right]$$

Thus the complete solution is $y = Ay_1 + By_2$ **Ans.**

BESSEL'S EQUATION

8.6 BESSEL'S EQUATION

The differential equation

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)\, y = 0$$

is called the *Bessel's differential equation*, and particular solutions of this equation are called Bessel's functions of order n.

8.7 SOLUTION OF BESSEL'S EQUATION

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)\, y = 0. \qquad ...(1)$$

Let $\quad y = \Sigma\, a_r x^{m+r}$ or $y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \ldots \qquad ...(2)$

so that $\quad \dfrac{dy}{dx} = \Sigma\, a_r (m+r)\, x^{m+r-1}$ and $\dfrac{d^2y}{dx^2} = \Sigma\, a_r (m+r)(m+r-1)\, x^{m+r-2}$

Substituting these values in the equation, we have

$$x^2 \Sigma\, a_r (m+r)(m+r-1)\, x^{m+r-2} + x\, \Sigma\, a_r (m+r)\, x^{m+r-1} + (x^2 - n^2)\, \Sigma\, a_r x^{m+r} = 0$$

or $\quad \Sigma\, a_r (m+r)(m+r-1)\, x^{m+r} + \Sigma\, a_r (m+r)\, x^{m+r} + \Sigma\, a_r x^{m+r+2} - n^2 \Sigma\, a_r x^{m+r} = 0$

or $\quad \Sigma\, a_r [(m+r)(m+r-1) + (m+r) - n^2]\, x^{m+r} + \Sigma\, a_r x^{m+r+2} = 0$

or $\quad \Sigma\, a_r [(m+r)^2 - n^2]\, x^{m+r} + \Sigma\, a_r x^{m+r+2} = 0.$

Contd…

Equating the coefficient of x^m to zero, we get

$$a_0\,[(m+0)^2 - n^2] = 0. \qquad (r = 0)$$

or $$m^2 = n^2 \text{ i.e. } m = n \qquad a_0 \neq 0$$

Equating the coefficient of x^{m+1} $\qquad r = 1$

$$a_1\,[(m+1)^2 - n^2] = 0 \text{ i.e. } a_1 = 0. \qquad \text{since } (m+1)^2 - n^2 \neq 0$$

Equating the coefficient of x^{m+r+2} to zero, to find relation in successive coefficients, we get

$$a_{r+2}\,[(m+r+2)^2 - n^2] + a_r = 0 \qquad \text{or} \qquad a_{r+2} = -\frac{1}{(m+r+2)^2 - n^2}\cdot a_r$$

Therefore $\qquad a_3 = a_5 = a_7 = \ldots = 0$, since $a_1 = 0$

If $r = 0$, $\qquad a_2 = -\dfrac{1}{(m+2)^2 - n^2}\,a_0$

If $r = 2$, $\qquad a_4 = -\dfrac{1}{(m+4)^2 - n^2}\,a_2 = \dfrac{1}{[(m+2)^2 - n^2]\,[(m+4)^2 - n^2)]}\,a_0$ $\qquad$ and so on

On substituting the values of the coefficients in (2) we have

$$y = a_0 x^m - \frac{a_0}{(m+2)^2 - n^2}\,x^{m+2} + \frac{a_0}{[(m+2)^2 - n^2]\,[(m+4)^2 - n^2]}\,x^{m+4} + \ldots$$

$$y = a_0\,x^m\left[1 - \frac{1}{(m+2)^2 - n^2}\,x^2 + \frac{1}{[(m+2)^2 - n^2]\,[(m+4)^2 - n^2]}\,x^4 - \ldots\right]$$

For $m = n$

$$y = a_0\,x^n\left[1 - \frac{1}{4\,(n+1)}\,x^2 + \frac{1}{4^2\cdot 2\,!\,(n+1)\,(n+2)}\,x^4 - \ldots\right]$$

where a_0 is an arbitrary constant.

For $m = -n$

$$y = a_0\,x^{-n}\left[1 - \frac{1}{4\,(-n+1)}\,x^2 + \frac{1}{4^2\cdot 2\,!\,(-n+1)\,(-n+2)}\,x^4 - \ldots\right]$$

BESSEL FUNCTION

JECRC UNIVERSITY
BUILD YOUR WORLD

8.8 BESSEL FUNCTIONS, $J_n(x)$

The Bessel's equation is $\quad x^2 \dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + (x^2 - n^2)\, y = 0$ ...(1)

Solution of (1) is

$$y = a_0 x^n \left[1 - \frac{x^2}{2 \cdot 2\,(n+1)} + \frac{x^4}{2.4.2^2\,(n+1)\,(n+2)} - \ldots + (-1)^r \frac{x^{2r}}{(2^r\, r\,!) \cdot 2^r\,(n+1)\,(n+2)\ldots(n+r)} + \ldots \right]$$

$$= a_0 x^n \sum_{r=0}^{\infty} (-1)^r \frac{x^{2r}}{2^{2r} \cdot r\,!\,(n+1)\,(n+2)\ldots(n+r)}$$

where a_0 is an arbitrary constant.

If $\quad a_0 = \dfrac{1}{2^n \Gamma(n+1)}$

The above solution is called Bessel's, function denoted by $J_n(x)$.

Thus $J_n(x) = \dfrac{1}{2^n \Gamma(n+1)} \Sigma\, (-1)^r \dfrac{x^{n+2r}}{2^{2r} \cdot r\,!\,(n+1)\,(n+2)\ldots(n+r)}$

$(\Gamma\, n+1 = n\,!)$

$$J_n(x) = \left(\frac{x}{2}\right)^n \left\{ \frac{1}{\Gamma(n+1)} - \frac{1}{\lfloor 1\, \Gamma(n+2)} \left(\frac{x}{2}\right)^2 + \frac{1}{\lfloor 2\, \Gamma(n+3)} \left(\frac{x}{2}\right)^4 - \frac{1}{\lfloor 3\, \Gamma(n+4)} \left(\frac{x}{2}\right)^6 + \ldots \right\}$$

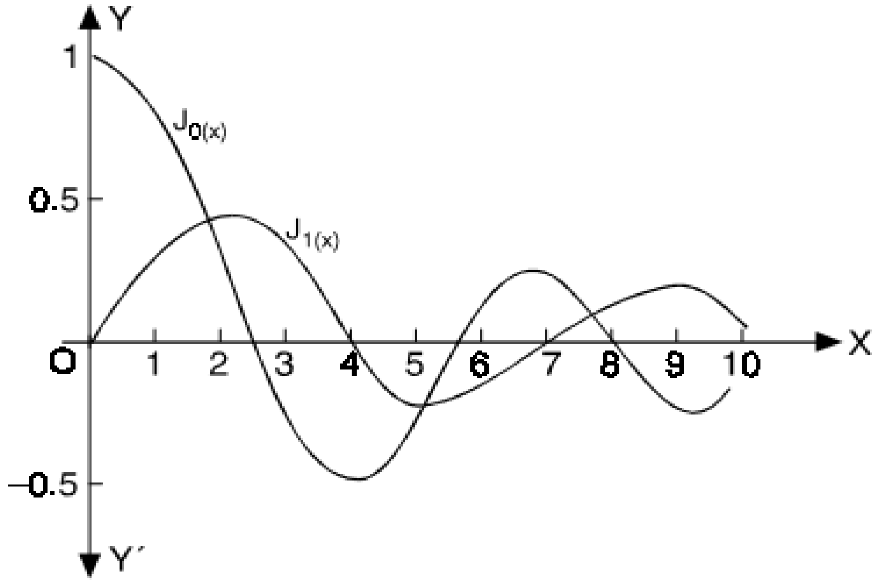
or $\quad J_n(x) = \displaystyle\sum_{r=0}^{\infty} \frac{(-1)^r}{r\,!\,(n+r)\,!} \left(\frac{x}{2}\right)^{n+2r}$ ...(2)

Contd…

If $n = 0, J_0(x) = \Sigma \frac{(-1)^r}{(r!)^2}\left(\frac{x}{2}\right)^{2r}$ or $J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \ldots$

If $n = 1$, $J_1(x) = \frac{x}{2} - \frac{x^3}{2^2 \cdot 4} + \frac{x^5}{2^2 \cdot 4^2 \cdot 6} - \ldots$

We draw the graphs of these two functions. Both the functions are oscillatory with a varying period and a decreasing amplitude.



Replacing n by $-n$ in (2), we get

$$J_{-n}(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!\,\Gamma(-n+r+1)}\left(\frac{x}{2}\right)^{-n+2r}$$

General Solution Of Bessel Equation

General solution of Bessel's Equation is

$$y = AJ_n(x) + BJ_{-n}(x)$$

Example 10. *Prove that*

$$J_{-n}(x) = (-1)^n J_n(x)$$

where n is a positive integer. *(A.M.I.E.T.E., Winter 2001)*

Solution.

$$J_{-n}(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{r!\,\Gamma(r-n+1)} \left(\frac{x}{2}\right)^{-n+2r}$$

$$= \sum_{r=0}^{n-1} \frac{(-1)^r \left(\frac{x}{2}\right)^{-n+2r}}{r!\,\Gamma(-n+r+1)} + \sum_{r=n}^{\infty} \frac{(-1)^r \left(\frac{x}{2}\right)^{-n+2r}}{r!\,\Gamma(-n+r+1)}$$

$$= 0 + \sum_{r=n}^{\infty} \frac{(-1)^r \left(\frac{x}{2}\right)^{-n+2r}}{r!\,\Gamma(-n+r+1)} \qquad \text{since } \Gamma(-\text{ve integer}) = \infty$$

On putting $r = n + k$

$$J_{-n}(x) = \sum_{k=0}^{\infty} \frac{(-1)^{n+k} \left(\frac{x}{2}\right)^{n+2k}}{(n+k)!\,\Gamma(k+1)}$$

$$= (-1)^n \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{x}{2}\right)^{n+2k}}{(n+k)!\,k!}$$

$$= (-1)^n J_n(x)$$

RECURRENCE FORMULAE

JECRC UNIVERSITY
BUILD YOUR WORLD

Formula I. $x J_n' = n J_n - x J_{n+1}$

Proof. We know that

$$J_n = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!\,\Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

Differentiating with respect to x, we get

$$J_n' = \sum \frac{(-1)^r (n+2r)}{r!\,\Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \frac{1}{2}.$$

$$x J_n' = n \sum \frac{(-1)^r}{r!\,\Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} + x \sum \frac{(-1)^r \cdot 2r}{2 \cdot r!\,\Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1}$$

$$= n J_n + x \sum_{r=1}^{\infty} \frac{(-1)^r}{(r-1)!\,\Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1}$$

$$= n J_n + x \sum_{s=0}^{\infty} \frac{(-1)^{s+1}}{s!\,\Gamma(n+s+2)} \left(\frac{x}{2}\right)^{n+2s+1} \qquad \text{Putting } r-1=s$$

$$= n J_n + x \sum_{s=0}^{\infty} \frac{(-1)^s}{s!\,\Gamma[(n+1)+s+1]} \left(\frac{x}{2}\right)^{(n+1)+2s}$$

$$= n J_n - x J_{n+1}$$ **Proved.**

Formula II. $x\,J_n' = -n\,J_n + x\,J_{n-1}$

Proof. We have $J_n = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!\,\Gamma(n+r+1)}\left(\frac{x}{2}\right)^{n+2r}$

Differentiating w.r.t. 'x' we get $J_n' = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{r!\,\Gamma(n+r+1)}\left(\frac{x}{2}\right)^{n+2r-1} \frac{1}{2}$

$$x\,J_n' = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{r!\,\Gamma(n+r+1)}\left(\frac{x}{2}\right)^{n+2r} = \sum_{r=0}^{\infty} \frac{(-1)^r [(2n+2r)-n]}{r!\,\Gamma(n+r+1)}\left(\frac{x}{2}\right)^{n+2r}$$

$$= \sum_{r=0}^{\infty} \frac{(-1)^r (2n+2r)}{r!\,\Gamma(n+r+1)}\left(\frac{x}{2}\right)^{n+2r} - n\sum_{r=0}^{\infty} \frac{(-1)^r}{r!\,\Gamma(n+r+1)}\left(\frac{x}{2}\right)^{n+2r}$$

$$= \sum_{r=0}^{\infty} \frac{(-1)^r\,2}{r!\,\Gamma(n+r)}\left(\frac{x}{2}\right)^{n+2r} - n\,J_n = x\sum_{r=0}^{\infty} \frac{(-1)^r}{r!\,\Gamma[(n-1)+r+1]}\left(\frac{x}{2}\right)^{(n-1+2r)} - n\,J_n$$

$$= x\,J_{n-1} - n\,J_n$$

Proved.

Contd…

Formula III. $2 J_n' = J_{n-1} - J_{n+1}$

Proof. We know that

$$x J_n' = n J_n - x J_{n+1} \quad \ldots(1) \qquad \text{Recurrence formula } I$$

$$x J_n' = -n J_n + x J_{n-1} \quad \ldots(2) \qquad \text{Recurrence formula } II$$

Adding (1) and (3), we get

$$2 x J_n' = -x J_{n+1} + x J_{n-1} \quad \text{or} \quad 2 J_n' = J_{n-1} - J_{n+1}$$

Proved.

Formula IV. $2 n J_n = x (J_{n-1} + J_{n+1})$

Proof. We know that

$$x J_n' = n J_n - x J_{n-1} \qquad \text{Recurrence formula I}$$

$$x J_n' = -n J_n + x J_{n+1} \qquad \text{Recurrence formula II}$$

Subtracting (2) from (1), we get

$$0 = 2 n J_n - x J_{n+1} - x J_{n-1} \quad \text{or} \quad 2 n J_n = x (J_{n-1} + J_{n+1})$$

Proved.

Formula V. $\dfrac{d}{dx}(x^{-n} \cdot J_n) = -x^{-n} J_{n+1}$

Proof. We know that

Contd…

$$x\,J_n' = n\,J_n - x\,J_{n+1}$$ Recurrence formula *I*

Multiplying by x^{-n-1}, we obtain

$$x^{-n}\,J_n' = n\,x^{-n-1}\,J_n - x^{-n}\,J_{n+1}$$

i.e., $$x^{-n}\,J_n' - n\,x^{-n-1}\,J_n = -\,x^{-n}\,J_{n+1}$$

or $$\frac{d}{dx}(x^{-n}\,J_n) = -\,x^{-n}\,J_{n+1}$$ **Proved.**

Formula VI. $\dfrac{d}{dx}(x^n\,J_n) = x^n\,J_{n-1}$

Proof. We know that

$$x\,J_n' = -\,n\,J_n + x\,J_{n-1}$$ Recurrence formula *II*

Multiplying by x^{n-1}, we have

$$x^n\,J_n' = -\,n\,x^{n-1}\,J_n + x^n\,J_{n-1}$$

i.e., $$x^n\,J_n' + n\,x^{n-1}\,J_n = x^n\,J_{n-1}$$

or $$\frac{d}{dx}(x^n\,J_n) = x^n\,J_{n-1}$$ **Proved.**

Example 12. *Prove that* $J_2'(x) = \left(1 - \frac{4}{x^2}\right) J_1(x) + \frac{2}{x} J_0(x)$ *where* $J_n(x)$ *is the Bessel function of first kind.* *(U.P. III Semester, Summer 2001)*

Solution. By recurrence formula *II*

$$x J_n' = -n J_n + x J_{n-1} \qquad ...(1)$$

On putting $n = 2$, in (1) we have $x J_2' = -2 J_2 + x J_1$

or
$$J_2' = -\frac{2}{x} J_2 + J_1 \qquad ...(2)$$

By recurrence formula *I*

$$x J_n' = n J_n - x J_{n+1} \qquad ...(3)$$

From (1) and (3) we have $-n J_n + x J_{n-1} = n J_n - x J_{n+1}$

On putting $n = 1$, $-J_1 + x J_0 = J_1 - x J_2$

or
$$-\frac{1}{x} J_1 + J_0 = \frac{1}{x} J_1 - J_2 \quad \text{or} \quad J_2 = \frac{2}{x} J_1 - J_0 \qquad ...(4)$$

Putting the value of J_2 from (4) in (2) we get

$$J_2' = -\frac{2}{x}\left(\frac{2}{x} J_1 - J_0\right) + J_1 = -\frac{4}{x^2} J_1 + \frac{2}{x} J_0 + J_1$$

$$= \left(1 - \frac{4}{x^2}\right) J_1 + \frac{2}{x} J_0$$

Proved.

8.17 LEGENDRE'S EQUATION

The differential equation $$(1 - x^2)\frac{d^2y}{dx^2} - 2\,x\,\frac{dy}{dx} + n\,(n+1)\,y = 0 \qquad \text{...(1)}$$

is known as Legendre's equation. The above equation can also be written as

$$\frac{d}{dx}\left\{(1 - x^2)\,\frac{dy}{dx}\right\} + n\,(n+1)\,y = 0 \qquad n\ \varepsilon\ I$$

This equation can be integrated in series of ascending or descending powers of x. *i.e.*, series in ascending or descending powers of x can be found which satisfy the equation (1).

Let the series in descending powers of x be

$$y = x^m\,(a_0 + a_1\,x^{-1} + a_2\,x^{-2} + \ldots) \qquad \text{...(2)}$$

or $$y = \sum_{r=0}^{\infty} a_r\,x^{m-r}$$

so that $$\frac{dy}{dx} = \sum_{r=0}^{\infty} a_r\,(m-r)^{m-r-1}$$

and $$\frac{d^2y}{dx^2} = \sum_{r=0}^{\infty} a_r\,(m-r)\,(m-r-1)\,x^{m-r-2}$$

Substituting these in (1), we have

Contd…

$$(1 - x^2) \sum_{r=0}^{\infty} a_r (m - r)(m - r - 1) x^{m - r - 2} - 2x \sum_{r=0}^{\infty} a_r (m - r) x^{m - r - 1} + n(n+1) \sum_{0}^{\infty} a_r x^{m - r} = 0$$

or $$\sum_{r=0}^{\infty} a_r (m - r)(m - r - 1) x^{m - r - 2} + \{ n(n+1) - 2(m - r) - (m - r)(m - r - 1) \} x^{m - r} a_r = 0$$

or $$\sum_{r=0}^{\infty} [(m - r)(m - r - 1) x^{m - r - 2} + \{ n(n+1) - (m - r)(m - r + 1) \} x^{m - r}] a_r = 0 \qquad \ldots(3)$$

The equation (3) is an identity and therefore coefficients of various powers of x must vanish. Now equating to zero the coefficients of x^m from the above we have ($r = 0$)

$$a_0 \{ n(n+1) - m(m+1) \} = 0$$

But $a_0 \neq 0$, as it is the coefficient of the very first term in the series.

Hence $n(n+1) - m(m+1) = 0 \qquad \ldots(4)$

i.e., $n^2 + n - m^2 - m = 0$ or $(n^2 - m^2) + (n - m) = 0$

or $(n - m)(n + m + 1) = 0$

which gives $m = n$ or $m = -n - 1 \qquad \ldots(5)$

This is important as it determines the index.

Next, equating to zero the coefficient of x^{m-1} by putting $r = 1$,

$a_1 [n(n+1) - (m - 1) m] = 0$

or $a_1 [(m + n)(m - n - 1)] = 0$

which gives $a_1 = 0 \qquad \ldots(6)$

Since $(m + n)(m - n - 1) \neq 0.$ by (5)

Again to find a relation in successive coefficients a_r, etc., equating the coefficient of $x^{m - r - 2}$ to zero, we get

Contd…

$$(m-r)(m-r-1)\,a_r + [n\,(n+1) - (m-r-2)(m-r-1)]\,a_{r+2} = 0$$

Now $n\,(n+1) - (m-r-2)(m-r-1) = n^2 + n - (m-r-1-1)(m-r-1)$

$$= -\,[(m-r-1)^2 - (m-r-1) - n^2 - n]$$

$$= -\,[(m-r-1+n)(m-r-1-n) - (m-r-1+n)]$$

$$= -\,[(m-r-1+n)(m-r-1-n-1)]$$

$$= (m-r+n-1)(m-r+n-2)$$

or $$(m-r)(m-r-1)\,a_r - (m-r+n-1)(m-r-n-2)\,a_{r+2} = 0$$

or $$a_{r+2} = \frac{(m-r)(m-r-1)}{(m-r+n-1)(m-r-n-2)}\,a_r \qquad \text{...(7)}$$

Now since $a_1 = a_3 = a_5 = a_7 = \ldots = 0$

For the two values given by (5) there arises following two cases.

Case I: When $m = n$

$$a_{r+2} = -\frac{(n-r)(n-r-1)}{(2n-r-1)(r+2)}\,a_r \qquad \text{from (7)}$$

so that, $$a_2 = -\frac{n\,(n-1)}{(2\,n-1)\,2}\,a_0,$$

$$a_4 = -\frac{(n-2)(n-3)}{(2\,n-3)\times 4}\,a_2 = \frac{n\,(n-1)(n-2)(n-3)}{(2\,n-1)(2\,n-3)\,2.4}\,a_0$$

and so on and $a_1 = a_3 = a_5 = \ldots = 0$

Contd…

Hence the series (2) becomes

$$y = a_0\left[x^n - \frac{n(n-1)}{(2n-1)\cdot 2}x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{(2n-1)(2n-3)\,2.4}\cdot x^{n-4} - \dots\right] \qquad \dots(8)$$

which is a solution of (1)

Case II: When $m = -(n+1)$, we have

$$a_{r+2} = \frac{(n+r+1)(n+r+2)}{(r+2)(2n+r+3)}a_r \qquad \text{from (7)}$$

so that

$$a_2 = \frac{(n+1)(n+2)}{2(2n+3)}a_0;$$

$$a_4 = \frac{(n+1)(n+2)(n+3)(n+4)}{2.4(2n+3)(2n+5)}a_0 \qquad \text{and so on.}$$

Hence the series (2) in this case becomes

$$y = a_0\left[x^{-n-1} + \frac{(n+1)(n+2)}{2\cdot(2n+3)}x^{-n-3} + \frac{(n+1)(n+2)(n+3)(n+4)}{2.4(2n+3)(2n+5)}\cdot x^{-n-5} + \dots\right] \qquad \dots(9)$$

This gives another solution of (1) in a series of descending powers of x.

Note. If we want to integrate the Legendre's equation in a series of ascending powers of x, we may proceed by taking

$$y = a_0 x^k + a_1 x^{k+1} + a_2 x^{k+2} + \dots = \sum_0^{\infty} a_r x^{k+r}$$

But integration in descending powers of x is more important than that in ascending powers of x.

8.18 LEGENDRE'S POLYNOMIAL $P_n(x)$.

Definition:

The Legendre's Equation is

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)\,y = 0 \qquad \ldots(1)$$

The solution of the above equation in the series of descending powers of x is

$$y = a_0\left[x^n - \frac{n(n-1)}{(2n-1)2}x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{(2n-1)(2n-3)2.4}x^{n-4}\ldots\right]$$

where a_0 is an arbitrary constant.

Now if n is a positive integer and $a_0 = \dfrac{1.3.5\ldots.(2n-1)}{n!}$ the above solution is $P_n(x)$, so that

$$P_n(x) = \frac{1.3.5\ldots.(2n-1)}{n!}\left[x^n - \frac{n(n-1)}{(2n-1)\cdot 2}x^{n-2} + \ldots\right]$$

Note 1. This is a terminating series.

When n is even, it contains $\frac{1}{2}n+1$ terms, the last term being

$$(-1)^{\frac{1}{2}n}\frac{n(n-1)(n-2)\ldots.1}{(2n-1)(2n-3)\ldots(n+1)\,2.4.6\ldots n}$$

And when n is odd it contains $\frac{1}{2}(n+1)$ terms and the last term in this case is

$$(-1)^{\frac{1}{2}(n-1)}\frac{n(n-1)\ldots 3.2}{(2n-1)(2n-3)\ldots(n+2)\,2.4\ldots(n-1)}x$$

$P_n(x)$ is called the *Legendre's functions of the first kind.*

Note. $P_n(x)$ is that solution of Legendre's equation (1) which is equal to unity when $x = 1$.

8.19 LEGENDRE'S FUNCTION OF THE SECOND KIND *i.e.* $Q_n(x)$.

Another solution of Legendre's equation

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0$$

when n is a positive integer

$$y = a_0\left[x^{-n-1} + \frac{(n+1)(n+2)}{2\cdot(2n+3)}x^{-n-3} + \ldots\right]$$

If we take $$a_0 = \frac{n!}{1.3.5\ldots(2n+1)}$$

the above solution is called $Q_n(x)$, so that

$$Q_n(x) = \frac{n!}{1.3.5.\ldots(2n+1)}\left[x^{-n-1} + \frac{(n+1)(n+2)}{2\cdot(2n+3)}x^{-n-3} + \ldots\right]$$

The series for $Q_n(x)$ is a non-terminating series.

8.20 GENERAL SOLUTION OF LEGENDRE'S EQUATION

Since $P_n(x)$ and $Q_n(x)$ are two independent solutions of Legendre's equation, therefore the most general solution of Legendre's equation is

$$y = AP_n(x) + BQ_n(x)$$

where A and B are two arbitrary constants.

Contd…

8.21 RODRIGUE'S FORMULA

$$P_n(x) = \frac{1}{2^n \cdot n!} \frac{d^n}{dx^n}(x^2-1)^n \qquad \text{(A.M.I.E.T.E., Winter 2001)}$$

Proof. Let

$$v = (x^2-1)^n \qquad \ldots(1)$$

Then

$$\frac{dv}{dx} = n(x^2-1)^{n-1}(2x)$$

Multiplying both sides by (x^2-1), we get

$$(x^2-1)\frac{dv}{dx} = 2n(x^2-1)^n x.$$

or

$$(x^2-1)\frac{dv}{dx} = 2nvx \qquad \ldots(2)$$

Now differentiating (2), $(n+1)$ times by Leibnitz's theorem, we have

$$(x^2-1)\frac{d^{n+2}v}{dx^{n+2}} + {}^{(n+1)}C_1(2x)\frac{d^{n+1}v}{dx^{n+1}} + {}^{(n+1)}C_2(2)\frac{d^nv}{dx^n} = 2n\left[x\frac{d^{n+1}v}{dx^{n+1}} + {}^{(n+1)}C_1(1)\frac{d^nv}{dx^n}\right]$$

or

$$(x^2-1)\frac{d^{n+2}v}{dx^{n+2}} + 2x\left[{}^{n+1}C_1 - n\right]\frac{d^{n+1}v}{dx^{n+1}} + 2\left[{}^{n+1}C_2 - n.{}^{(n+1)}C_1\right]\frac{d^nv}{dx^n} = 0$$

or

$$(x^2-1)\frac{d^{n+2}v}{dx^{n+2}} + 2x\frac{d^{n+1}v}{dx^{n+1}} - n(n+1)\frac{d^nv}{dx^n} = 0 \qquad \ldots(3)$$

If we put $\frac{d^nv}{dx^n} = y$, (3) becomes

$$(x^2-1)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} - n(n+1)y = 0$$

or

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0$$

Contd…

This shows that $y = \frac{d^n v}{dx^n}$ is a solution of Legendre's equation.

$$\therefore \qquad C \frac{d^n v}{dx^n} = P_n(x) \qquad \ldots(4)$$

where C is a constant.

But $\qquad v = (x^2 - 1)^n = (x+1)^n (x-1)^n$

so that
$$\frac{d^n v}{dx^n} = (x+1)^n \frac{d^n}{dx^n}(x-1)^n + {}^n C_1 \cdot n(x+1)^{n-1} \cdot \frac{d^{n-1}}{dx^{n-1}}(x-1)^n + \ldots + (x-1)^n \frac{d^n}{dx^n}(x+1)^n = 0$$

when $x = 1$, $\qquad \frac{d^n v}{dx^n} = 2^n \cdot n\,!$

All the other terms disappear as $(x-1)$ is a factor in every term except first.

Therefore when $x = 1$, (4) gives

$$C \cdot 2^n \cdot n\,! = P_n(1) = 1 \qquad\qquad P_n(1) = 1$$

$$C = \frac{1}{2^n \cdot n\,!}. \qquad \ldots(5)$$

Substituting the value of C from (1) in (5) we have

$$P_n(x) = \frac{1}{2^n \cdot n\,!} \frac{d^n v}{dx^n}$$

$$P_n(x) = \frac{1}{2^n \lfloor n} \frac{d^n}{dx^n}(x^2 - 1)^n$$

Contd…

Example 27. *Let $P_n(x)$ be the Legendre polynomial of degree n. Show that for any function, $f(x)$, for which the nth derivative is continuous,*

$$\int_{-1}^{1} f(x)\, P_n(x)\, dx = \frac{(-1)^n}{2^n\, n!} \int_{-1}^{1} (x^2-1)^n f^n(x)\, dx.$$

Solution. $\int_{-1}^{1} f(x)\, P_n(x)\, dx = \int_{-1}^{+1} f(x) \cdot \frac{1}{2^n \lfloor n} \frac{d^n}{dx^n}(x^2-1)^n\, dx$

$$\left[P_n(x) = \frac{1}{2^n \lfloor n} \frac{d^n}{dx^n}(x^2-1)^n \right]$$

$$= \frac{1}{2^n \lfloor n} \int_{-1}^{+1} f(x) \cdot \frac{d^n}{dx^n}(x^2-1)^n\, dx$$

Integrating by parts, we get

$$= \frac{1}{2^n \lfloor n} \left[f(x) \cdot \frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n - \int f'(x) \cdot \frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n\, dx \right]_{-1}^{+1}$$

$$= \frac{1}{2^n \lfloor n} \left[0 - \int_{-1}^{+1} f'(x) \cdot \frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n\, dx \right]$$

$$= \frac{(-1)}{2^n \lfloor n} \int_{-1}^{+1} f'(x) \cdot \frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n\, dx$$

Again integrating by parts, we have

$$= \frac{(-1)}{2^n \lfloor n} \left[f'(x) \frac{d^{n-2}}{dx^{n-2}}(x^2-1)^n - \int f''(x) \cdot \frac{d^{n-2}}{dx^{n-2}}(x^2-1)^n\, dx \right]_{-1}^{+1}$$

$$= \frac{(-1)^2}{2^n \lfloor n} \int_{-1}^{+1} f''(x) \frac{d^{n-2}}{dx^{n-2}} \cdot (x^2-1)^n\, dx$$

Integrating $(n-2)$ times, by parts, we get

$$= \frac{(-1)^n}{2^n \lfloor n} \int_{-1}^{+1} f^n(x)\, (x^2-1)^n\, dx$$

Proved.

Thank You